UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement MOM Capacitors Monte Carlo Mismatch SPICE Model Document

Version 0.2 Phase 1
2010/08/03

Table of Contents

List of Figures

List of Tables

1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2010/08/03	Eddie Chang	1. Original release. 2. This Monte Carlo model includes both process and mis-matching characteristics extracted from model cards as below： L110AE_MOM_V021.mdl L110AE_MOM_V021.lib

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using UMC's L110E Mix-Mode/RF Process.

1.3 Model Usage

1.3.1 Simulators

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor Graphics ELDO version 2008.1 (6.11_1.1)

*Note: Known difference among different simulators:
Since HSPICE and ELDO don't directly support voltage coefficients in capacitor models, there is no voltage coefficients included in HSPICE and ELDO models. Spectre model has included voltage coefficients for MOM capacitors.

1.3.2 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. User can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

mc: Monte-Carlo model
tt: Typical MOM capacitor case
ss : Maximum MOM capacitor case
ff: Minimum MOM capacitor case

1.3.3 Mismatching parameters in corner model

The mismatching parameters are included in the corner models. Therefore, user can run Monte-Carlo mismatching simulation over a corner model. For example, user can evaluate the mismatching performance at the max case.

1.3.4 Sigma number for Monte-Carlo model

User can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

1.3.5 Model usage

The following is the definition of sub-circuit model parameters list:

c: capacitance of mom capacitor
l: length of mom capacitor
mh: number of unit cells
nf: number of fingers
nm: number of metal layers
bm : bottom layer
shielding_flag : 1 for with shielding layer structure
0 for without shielding layer structure

metal_ring_flag : 1 for with metal ring structure
0 for without metal ring structure
presim_flag : 1 for previous simulation
0 for post simulation
mis_flag: the flag parameter is used to turn on/off mismatching parameters for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

Follow these steps to invoke the model file during simulation.

-Define the flag of Monte-Carlo simulation.
PROCESS: 1 to turn on the MC process simulation
0 to turn off the MC process simulation
Example: .param process=1
.param process=0
-Define sigma value for MC simulation.
Example: .param sigma=3
-Define the number of Monte Carlo iterations.
Example: sweep monte=200 (for HSPICE and ELDO)
montecarlo numruns=200 (for SPECTRE)

The instant parameters for MOM capacitance Monte Carlo simulation could be c, nf, nm or l (before-shrink), nf, nm. It is meaningless if instant parameters c and l are used at the same time.

For Synopsys HSPICE

-Copy model files into your simulation directory.

-Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

.lib "./ModelFileName_MC_CORNER.lib" CaseName

, where CaseName could be MC, TT, SS, FF.

-Defince device condition by "Xxx" statement.

Example:

For non-mesh symmetry MOM capacitors

Non-shielding layer

Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

or

Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

With shielding layer

Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

or

Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh symmetry MOM capacitors

Non-shielding layer

Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0

or

Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0

With shielding layer

Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1

or

Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1

For non-mesh asymmetry MOM capacitors
Xasymom Vtop Vbottom Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 presim_flag=1 metal_ring_flag=1 or 0
or
Xasymom Vtop Vbottom Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh asymmetry MOM capacitors
Xasymom Vtop1 Vbottom1 Vtop2 Vbottom2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 presim_flag=1
or
Xasymom Vtop1 Vbottom1 Vtop2 Vbottom2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 presim_flag=1

For non-mesh stripe symmetry MOM capacitors
Non-shielding layer
Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
or
Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
With shielding layer
Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0
or
Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh stripe symmetry MOM capacitors
Non-shielding layer
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0
or
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0
With shielding layer
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1
or
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1

For Cadence SPECTRE

-Copy model files into your simulation directory.

-Add the following statement in the input file to the simulator for 90% dimension shrinkage.

SetOption 1 options scale=0.9

include "./ModelFileName_MC_CORNER.lib.scs" section=CaseName

, where CaseName could be mc, tt, ss, ff.

-Defince device condition by "Xxx" statement.

Example:

For non-mesh symmetry MOM capacitors

Non-shielding layer

Xsymom (Vport1 Vport2 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

or

Xsymom (Vport1 Vport2 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

With shielding layer

Xsymom (Vport1 Vport2 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

or

Xsymom (Vport1 Vport2 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh symmetry MOM capacitors

Non-shielding layer

Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0

or

Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0

With shielding layer

Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1

or

Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1

For non-mesh asymmetry MOM capacitors

Xasymom (Vtop Vbottom Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=1 presim_flag=1 metal_ring_flag=1 or 0

or

Xasymom (Vport1 Vport2 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh asymmetry MOM capacitors

Xasymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=1 presim_flag=1
or
Xasymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=1 presim_flag=1

For non-mesh stripe symmetry MOM capacitors
Non-shielding layer
Xsymom (Vport1 Vport2 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
or
Xsymom (Vport1 Vport2 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
With shielding layer
Xsymom (Vport1 Vport2 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0
or
Xsymom (Vport1 Vport2 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh stripe symmetry MOM capacitors
Non-shielding layer
Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0
or
Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0
With shielding layer
Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1
or
Xsymom (Vport1 Vport2 Vport3 Vport4 Vwell) DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1

For Mentor Graphics ELDO
-Copy following files into your simulation directory.
-Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "./ModelFileName_MC_CORNERlib.eldo" CaseName
, where CaseName could be MC, TT, SS, FF.
-Defince device condition by "Xxx" statement.
Example:
For non-mesh symmetry MOM capacitors
Non-shielding layer

Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
or
Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
With shielding layer
Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0
or
Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh symmetry MOM capacitors
Non-shielding layer
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0
or
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0
With shielding layer
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1
or
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1

For non-mesh asymmetry MOM capacitors
Xasymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 presim_flag=1 metal_ring_flag=1 or 0
or
Xasymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh asymmetry MOM capacitors
Xasymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 presim_flag=1
or
Xasymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 presim_flag=1

For non-mesh stripe symmetry MOM capacitors
Non-shielding layer
Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
or

Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0
With shielding layer
Xsymom Vport1 Vport2 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0
or
Xsymom Vport1 Vport2 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh stripe symmetry MOM capacitors
Non-shielding layer
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=1 shielding_flag=0
or
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=1 shielding_flag=0
With shielding layer
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName c=100f mh=2 nf=14 nm=6 bm=2 shielding_flag=1
or
Xsymom Vport1 Vport2 Vport3 Vport4 Vwell DeviceName l=3u mh=2 nf=14 nm=6 bm=2 shielding_flag=1

The available DeviceName in this model are:

Symmetric type MOM Capacitor: momcaps_sy_ mml110ae
Asymmetric type MOM Capacitor: momcaps_as_ mml110ae
Stripe symmetric type MOM Capacitor: momcaps_stripe_ mml110ae
Symmetric type Mesh MOM Capacitor: momcaps_symesh_ mml110ae
Asymmetric type Mesh MOM Capacitor: momcaps_asmesh_ mml110ae
Stripe symmetric type Mesh MOM Capacitor: momcaps_stripemesh_ mml110ae

Temperature range: -55℃ ~ +125 ℃

This model is valid within the following model scaling range. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Before shrinkage

Table 1-2 Model scaling range for Symmetric capacitors

Parameter		Unit	Min	Max	Limitation
l		um	4.4	78.4	
nf		integer	14	198	Even number
nm		integer	3	6	
w		um	Fixed at 0.2		
s		um	Fixed at 0.2		
c	nm=6	F	37.09f	9.58p	
	nm=5	F	30.86f	7.973p	
	nm=4	F	24.51f	6.334p	
	nm=3	F	17.98f	4.64p	

Table 1-3 Model scaling range for Asymmetric MOM capacitors

Parameter		Unit	Min	Max	Limitation
l		um	4.4	78.4	
nf		integer	15	199	odd number
nm		integer	4	6	
w		um	Fixed at 0.2		
s		um	Fixed at 0.2		
c	nm=6	F	37.79f	8.463p	
	nm=5	F	30.69f	6.872p	
	nm=4	F	23.58f	5.28p	

Table 1-4 Model scaling range for stripe symmetric MOM capacitors

Parameter		Unit	Min	Max	Limitation
l		um	4.4	78.4	
nf		integer	14	198	Even number
nm		integer	3	6	
w		um	Fixed at 0.2		
s		um	Fixed at 0.2		
c	nm=6	F	37.09f	9.58p	
	nm=5	F	30.86f	7.973p	
	nm=4	F	24.51f	6.334p	
	nm=3	F	17.98f	4.64p	

2 Monte Carlo DC Simulation Results for Process Variation

Five hundred (500) Monte Carlo iteration performed in the following simulation conditions, and the following corner points show TT,FF and SS values, individually

A reasonable number of Monte Carlo iteration is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)-1/2.

2.1 Simulation Result Plots

The following figures show the Monte Carlo simulation results of symmetric, stripe and asymmetric MOM capacitors (C=35.95fF, nf=14, nm=6, bm=1; C=30.72fF, nf=14, nm=5, bm=2 and C=36.592fF, nf=15, nm=6, bm=1) at 25 ℃.

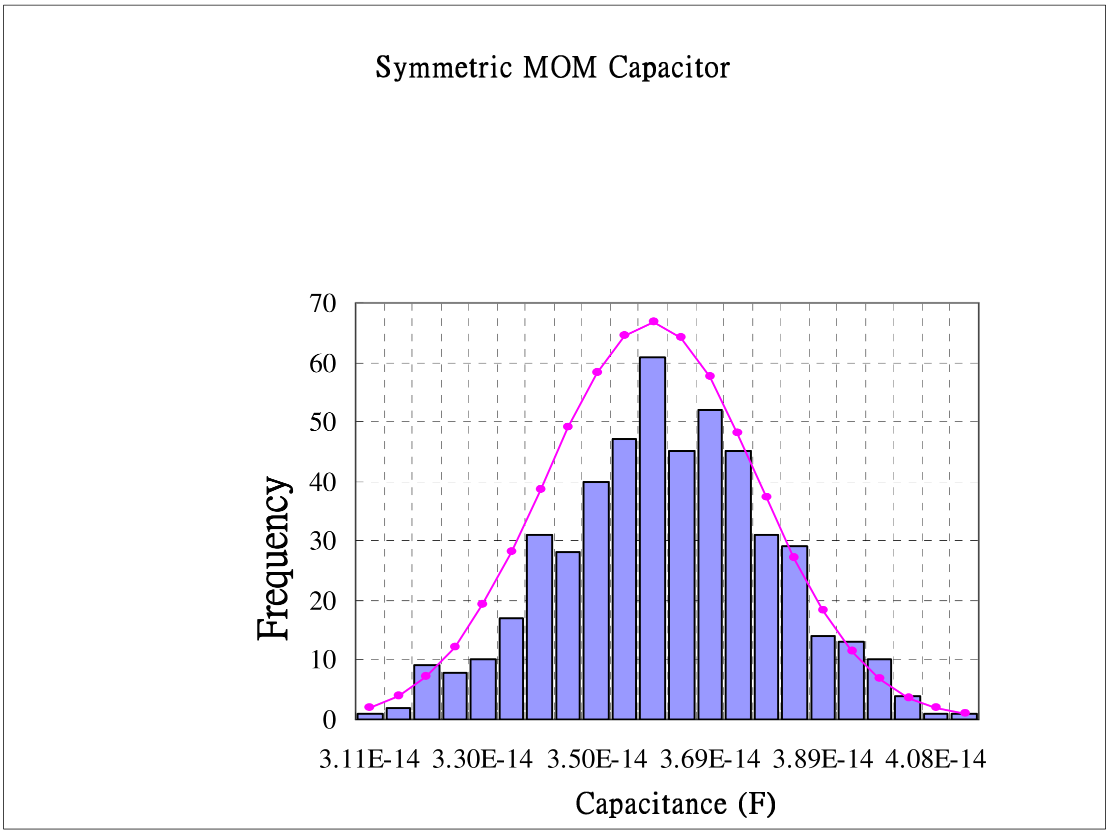

Figure 2-1 Monte Carlo plots of Symmetric MOM Capacitor

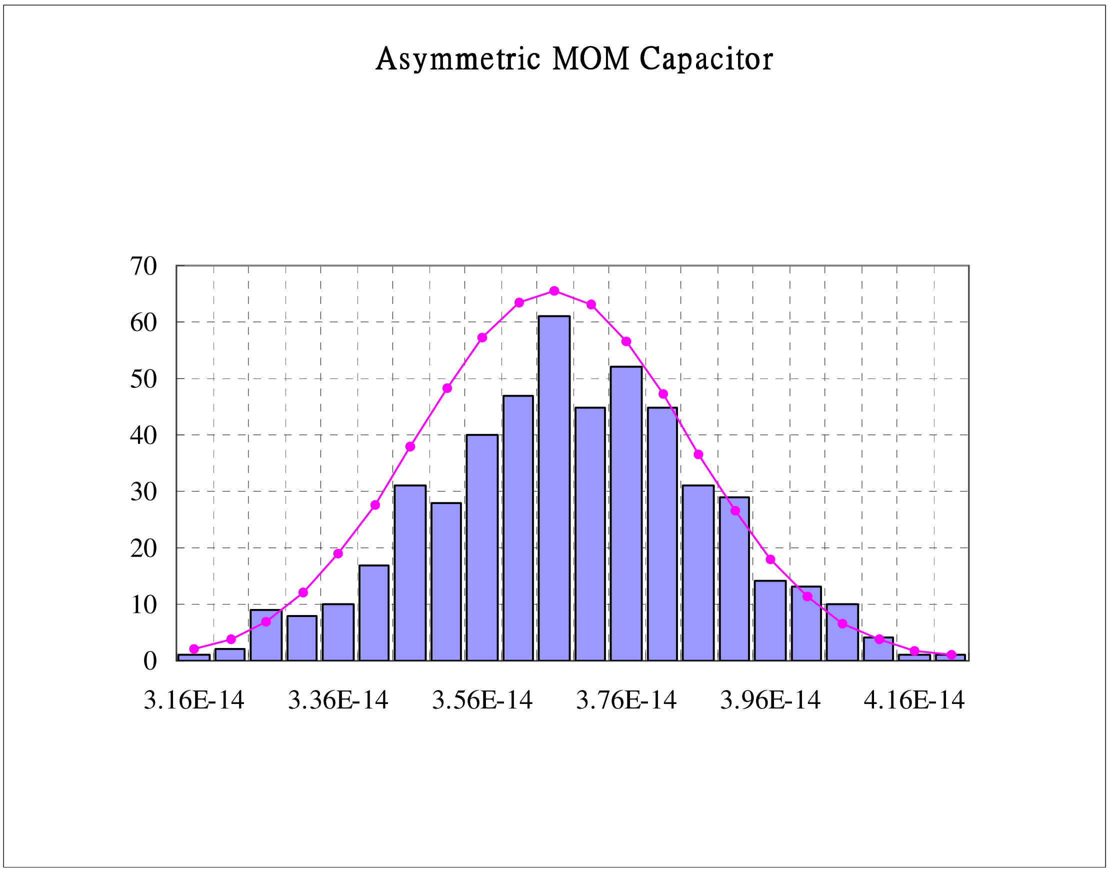

Figure 2-2 Monte Carlo plots of Asymmetric MOM Capacitor

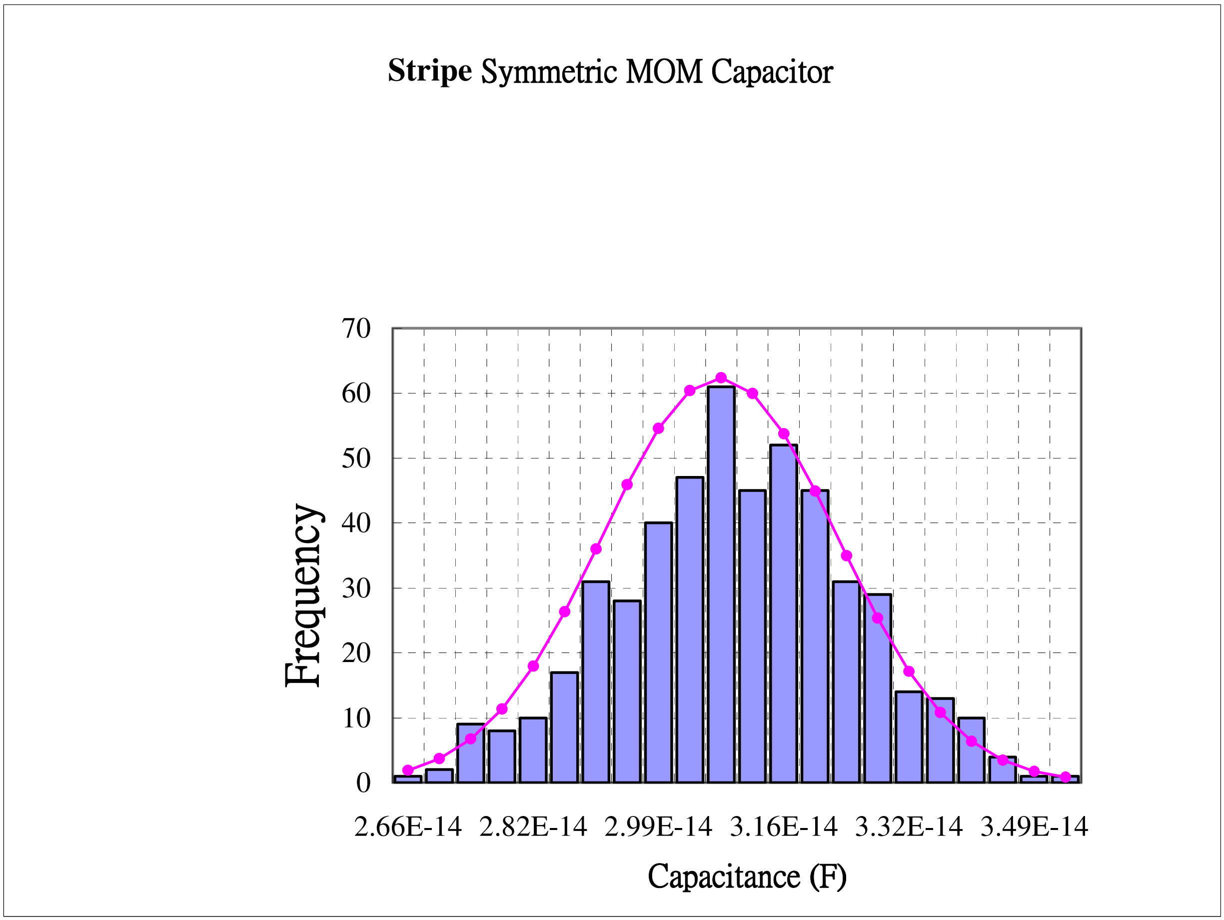

Figure 2-3 Monte Carlo plots of Stripe MOM Capacitor

The following figures show the Monte Carlo simulation results of symmetric, stripe and asymmetric mesh MOM capacitors (C=9317.17fF, mh=256, nf=14, nm=6, bm=1; C=7874.4fF, mh=256, nf=14, nm=5, bm=2 and C=2370.68fF, mh=64, nf=15, nm=6, bm=1) at 25 ℃.

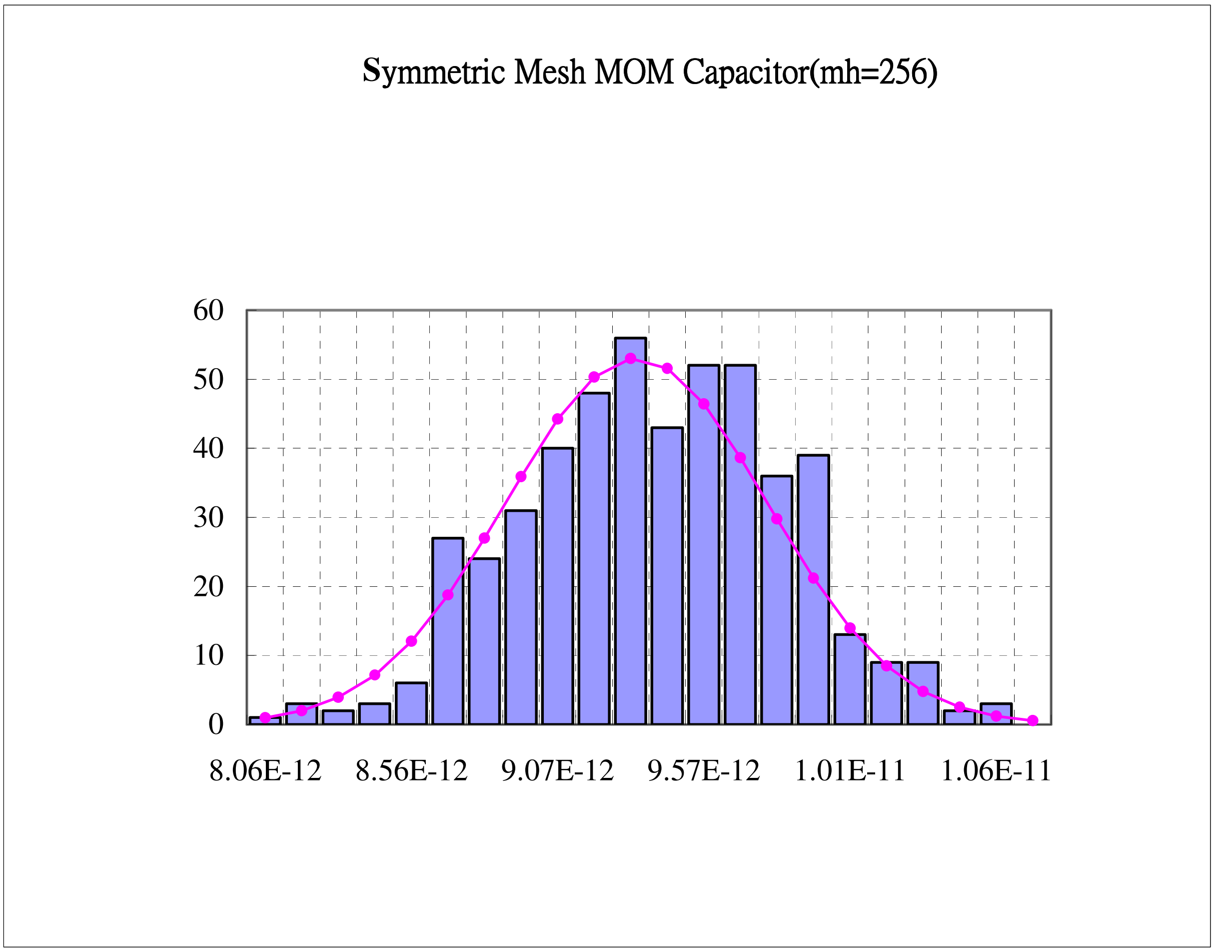

Figure 2-4 Monte Carlo plots of Symmetric Mesh MOM Capacitor

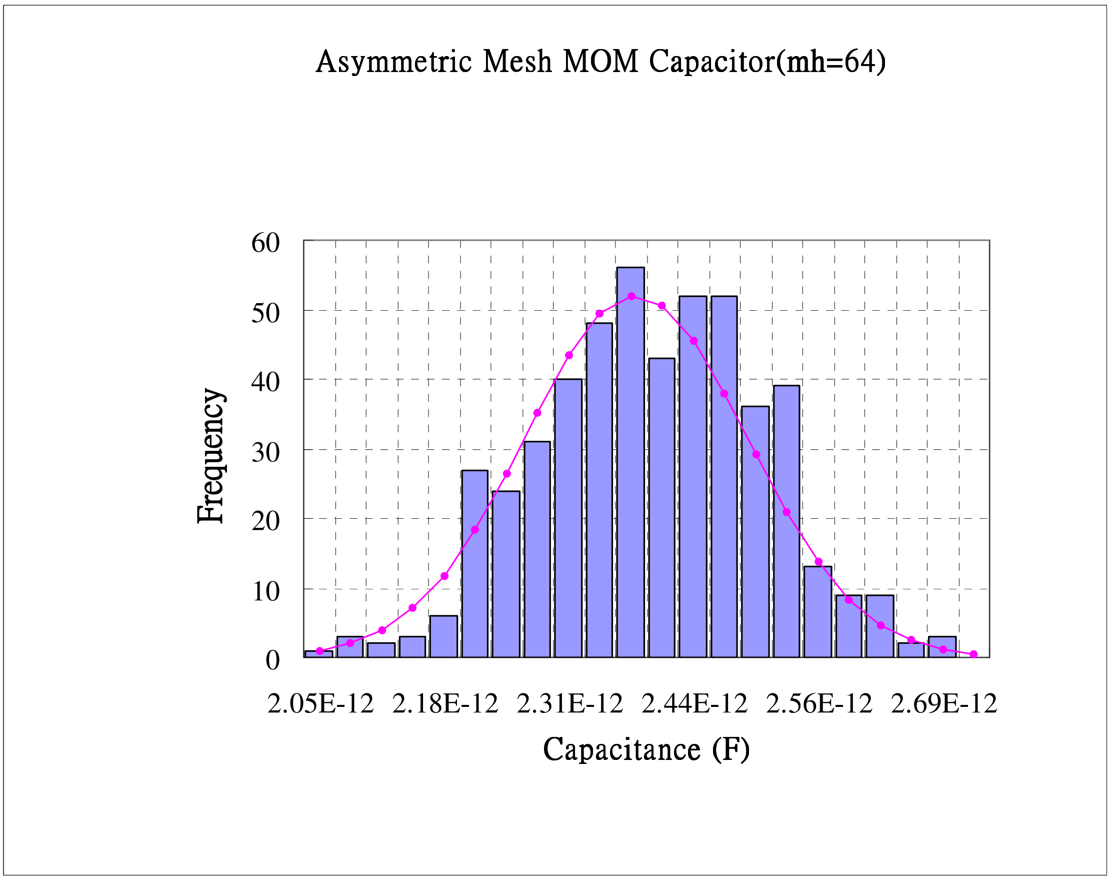

Figure 2-5 Monte Carlo plots of Asymmetric Mesh MOM Capacitor

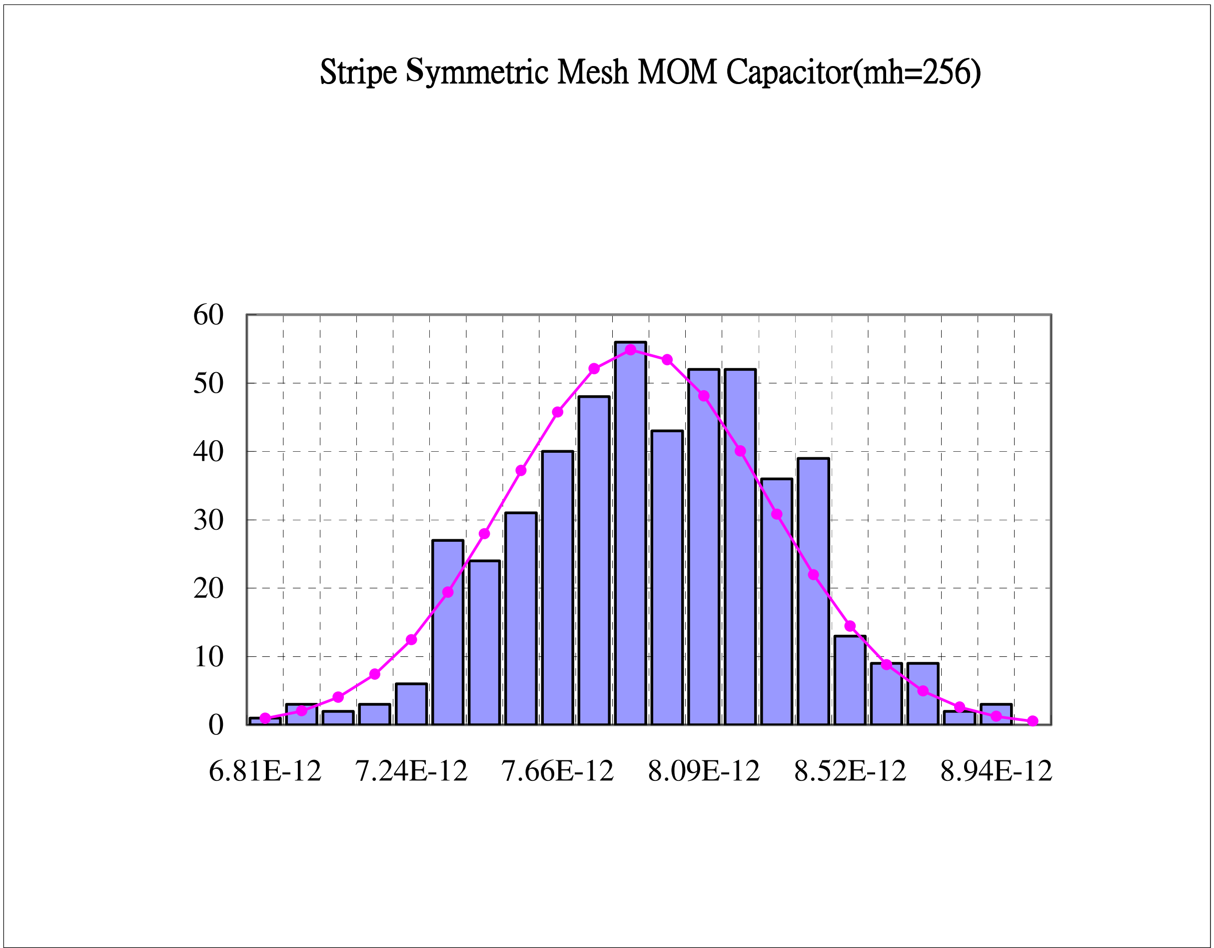

Figure 2-6 Monte Carlo plots of Stripe Symmetric Mesh MOM Capacitor

3 Monte Carlo Simulation Results for Mismatch

3.1 Simulation Result Plots

The following figures show the Monte Carlo mis-matching simulation results of Symmetric MOM capacitor at 25 ℃.

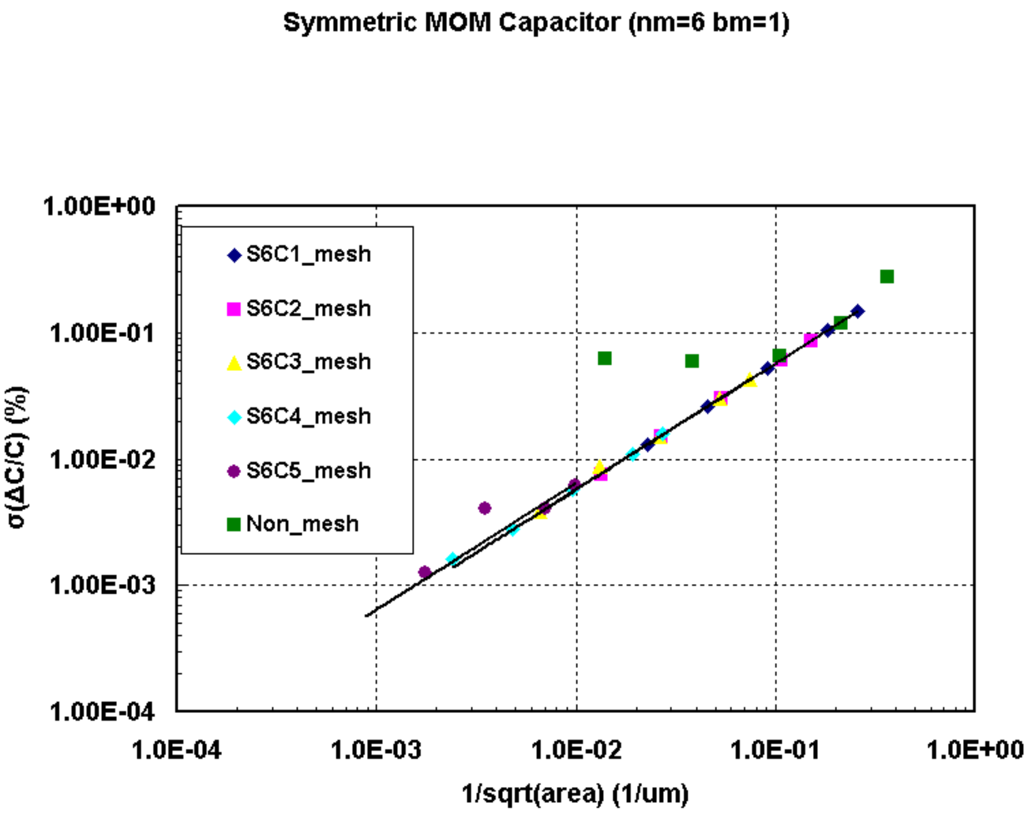

Figure 3-1 Mis-matching simulation of Symmetric MOM capacitor (nm=6 bm=1)

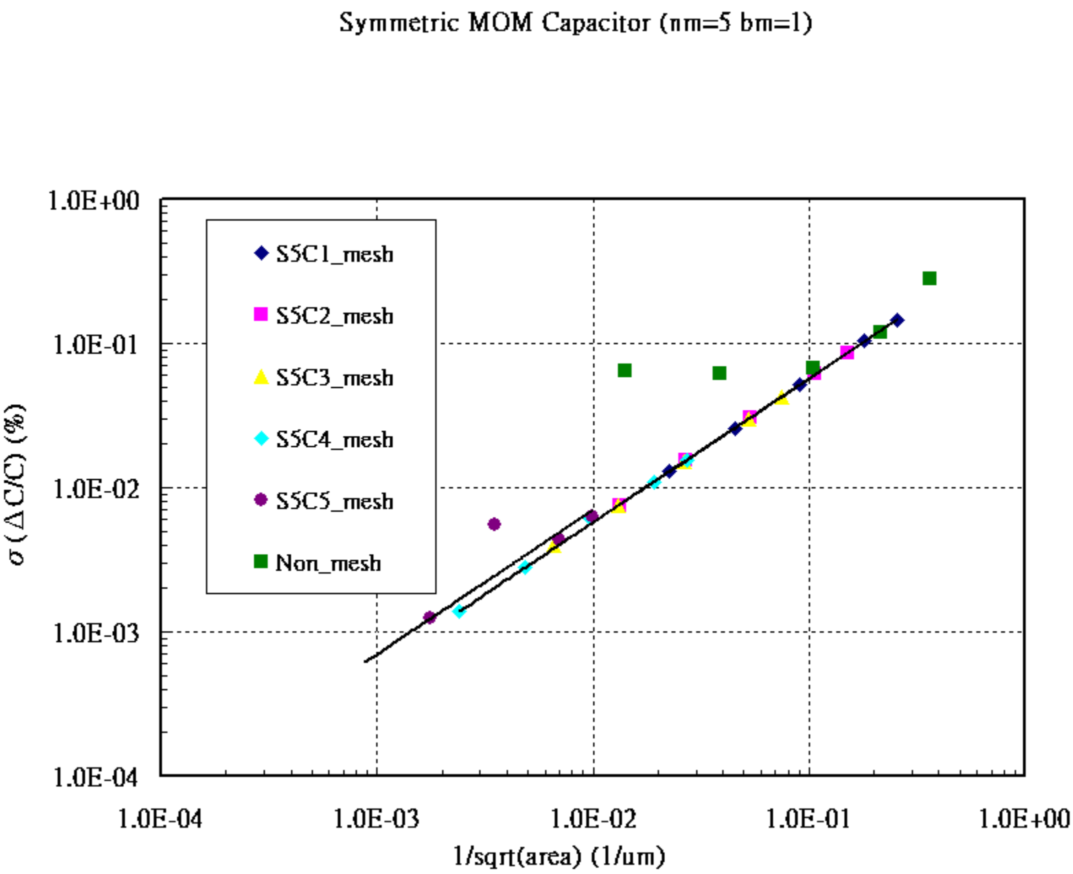

Figure 3-2 Mis-matching simulation of Symmetric MOM capacitor (nm=5 bm=1)

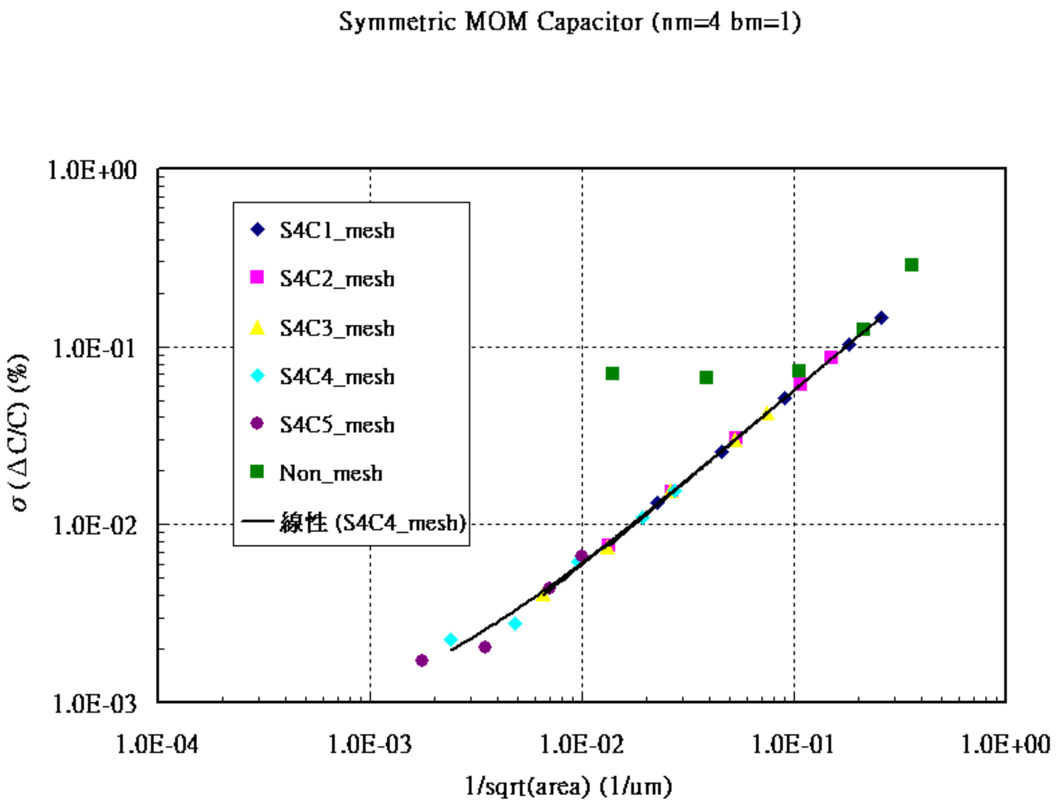

Figure 3-3 Mis-matching simulation of Symmetric MOM capacitor (nm=4 bm=1)

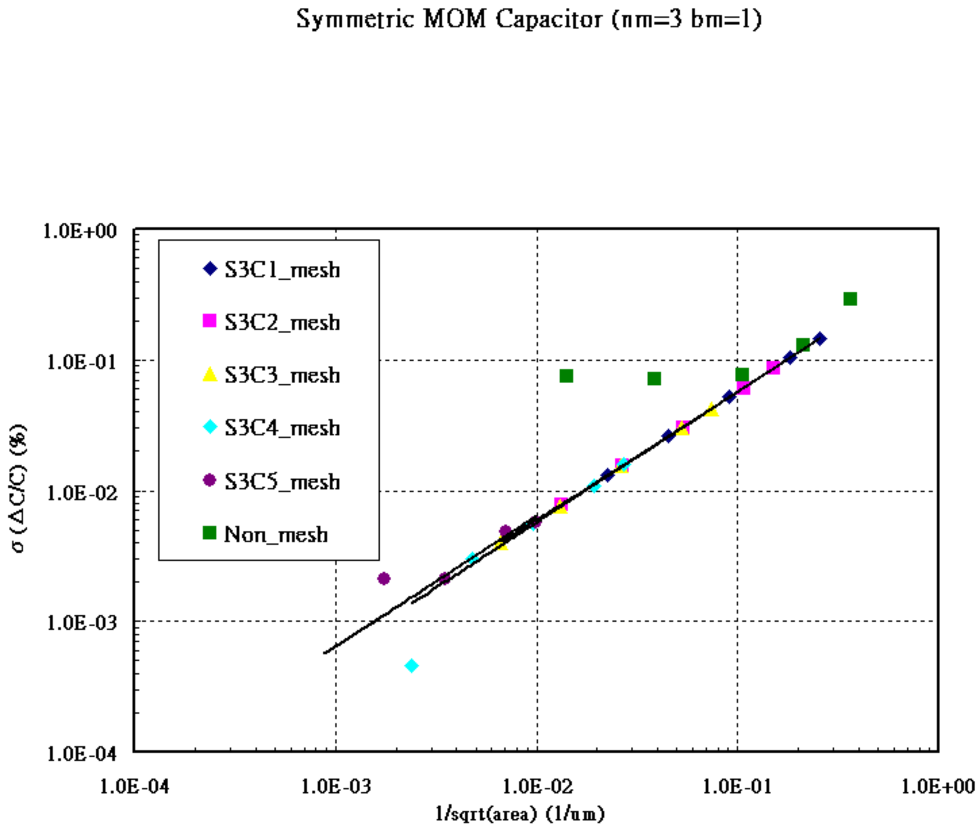

Figure 3-4 Mis-matching simulation of Symmetric MOM capacitor (nm=3 bm=1)

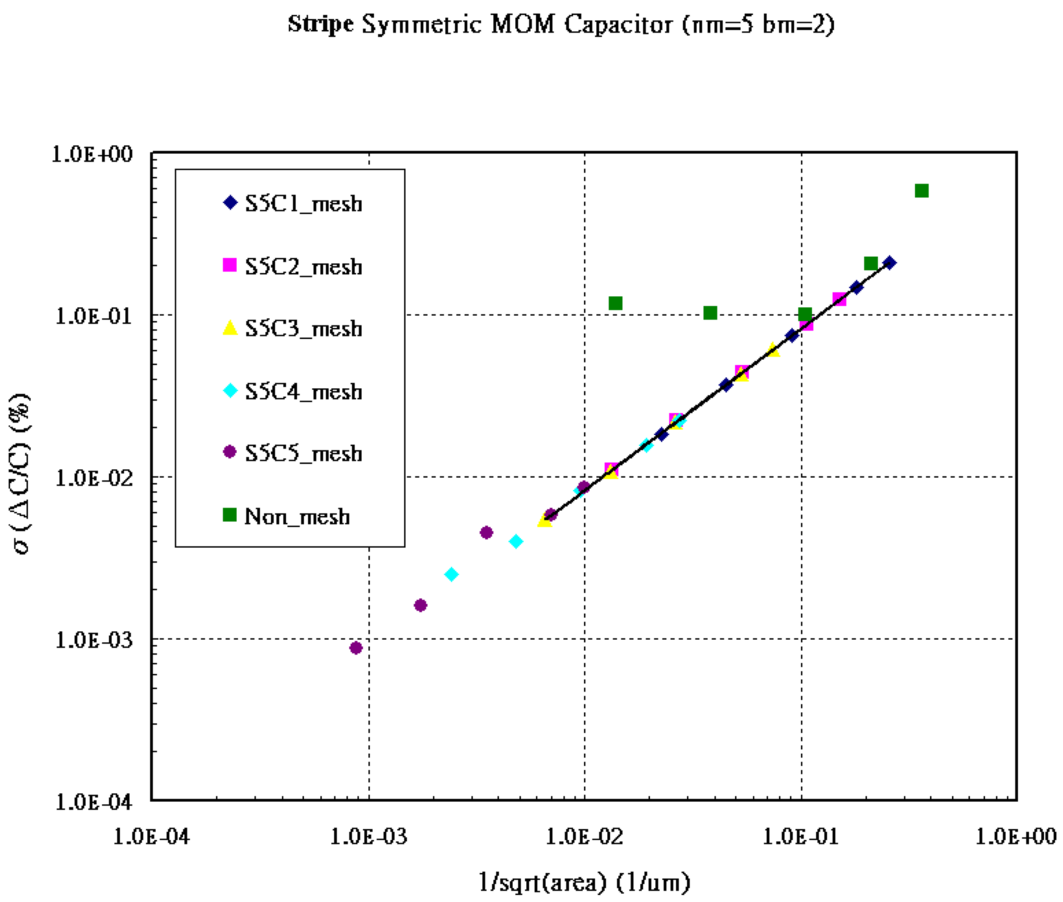

Figure 3-5 Mis-matching simulation of Symmetric MOM capacitor (nm=5 bm=2)

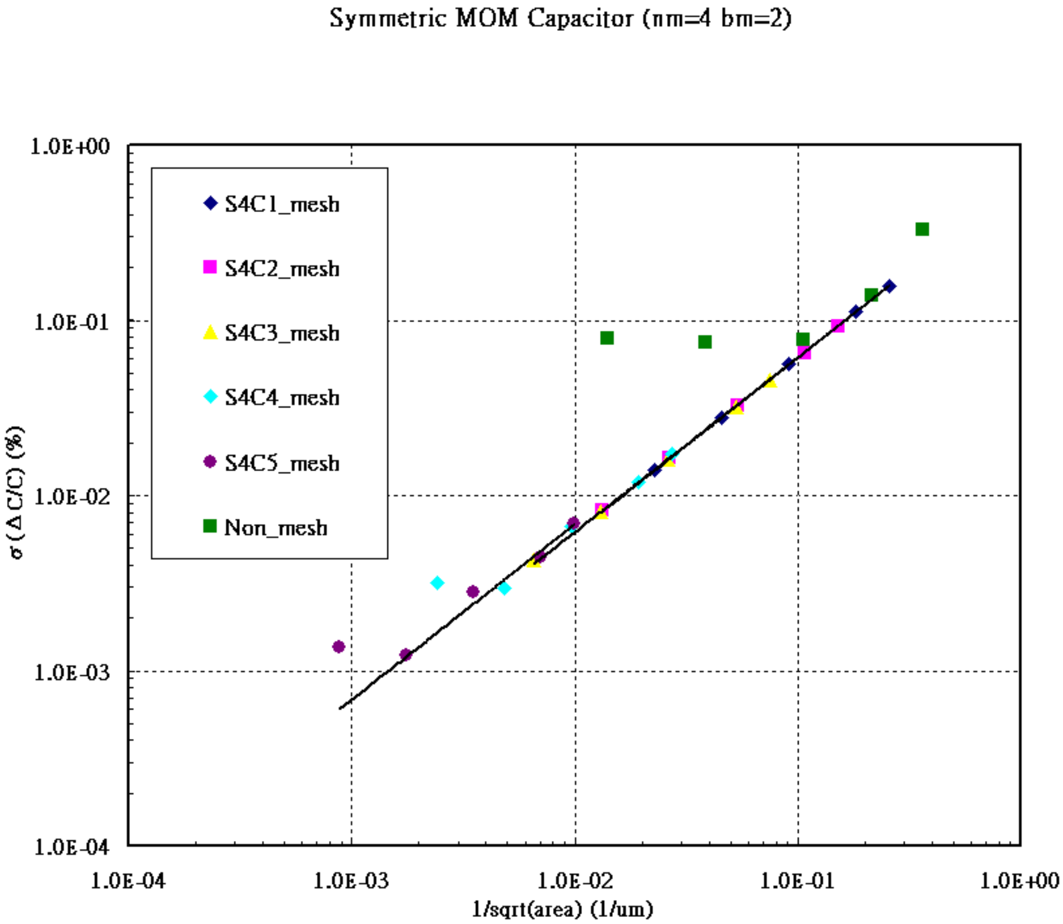

Figure 3-6 Mis-matching simulation of Symmetric MOM capacitor (nm=4 bm=2)

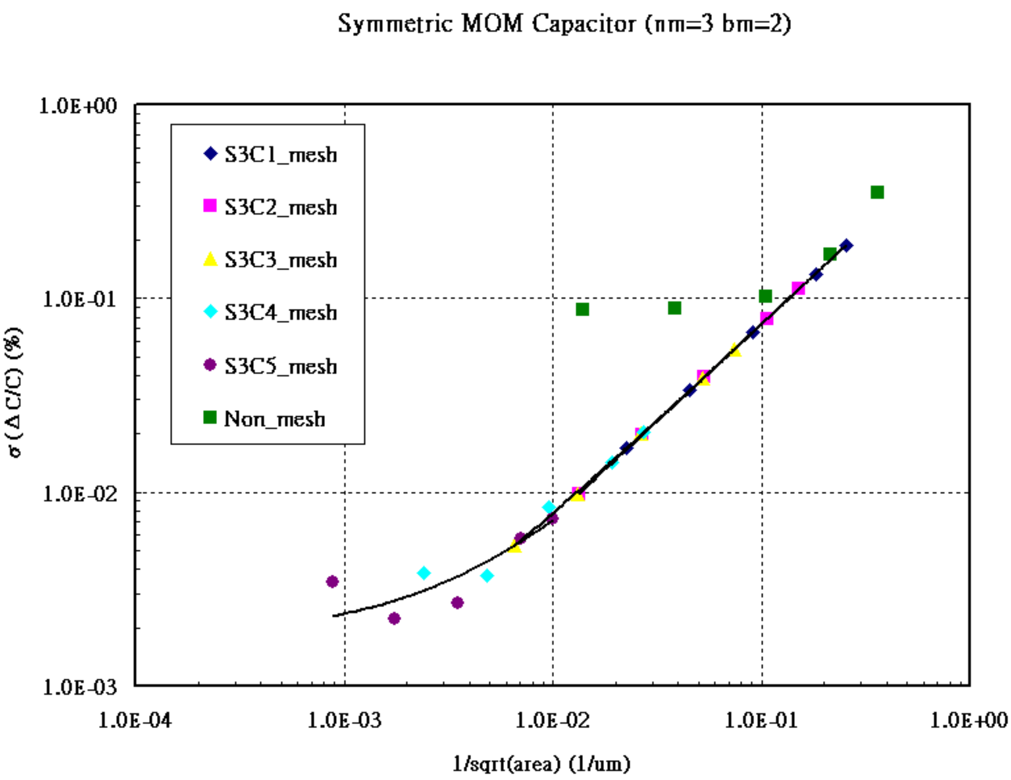

Figure 3-7 Mis-matching simulation of Symmetric MOM capacitor (nm=3 bm=2)

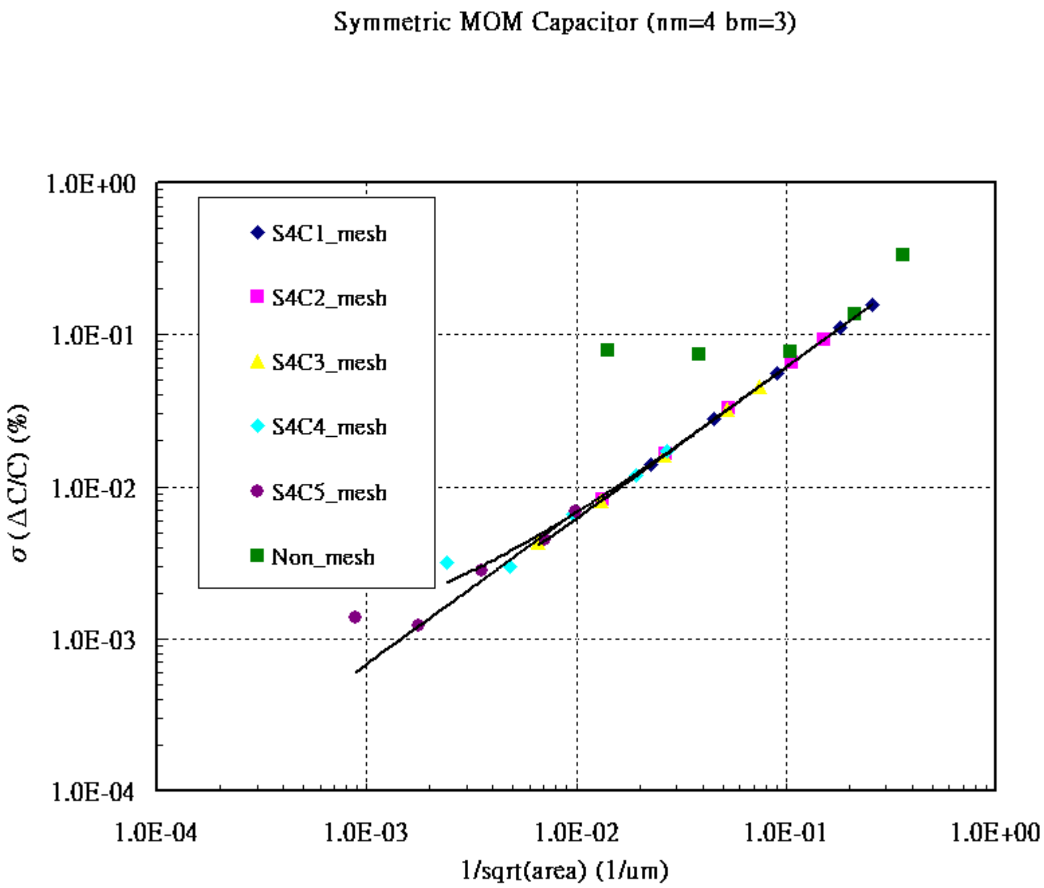

Figure 3-8 Mis-matching simulation of Symmetric MOM capacitor (nm=4 bm=3)

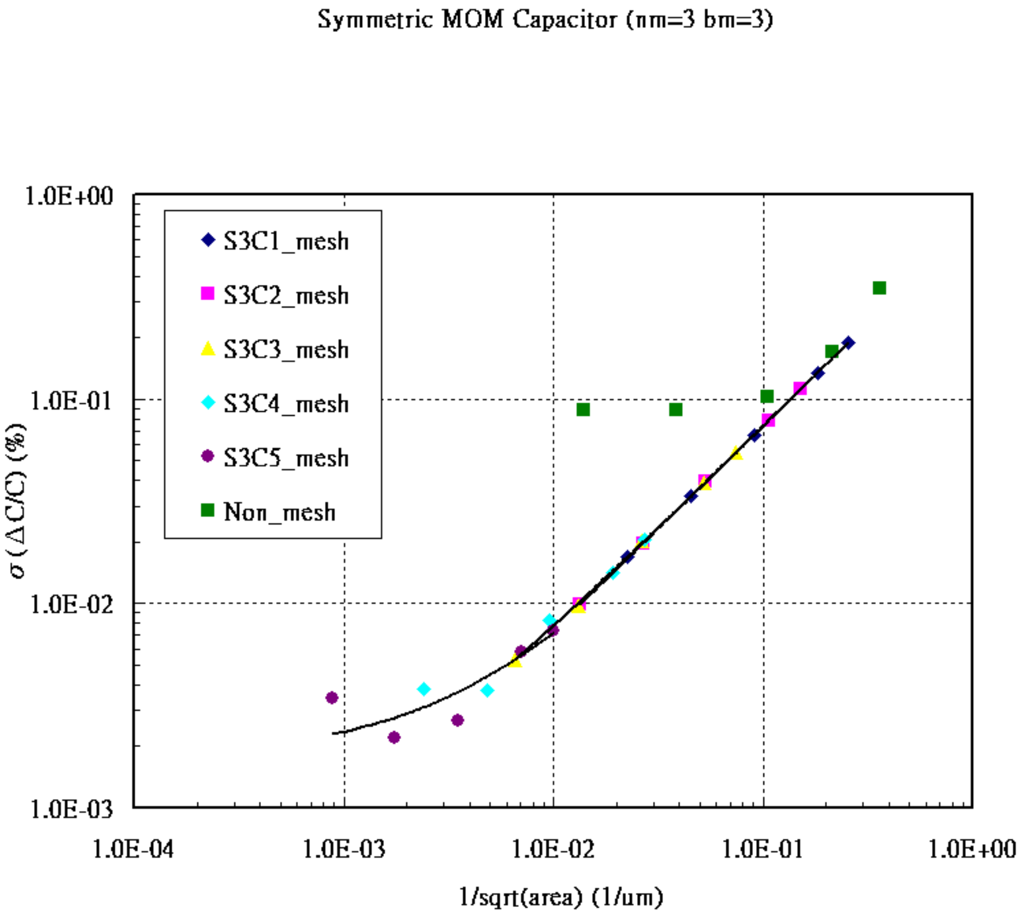

Figure 3-9 Mis-matching simulation of Symmetric MOM capacitor (nm=3 bm=3)

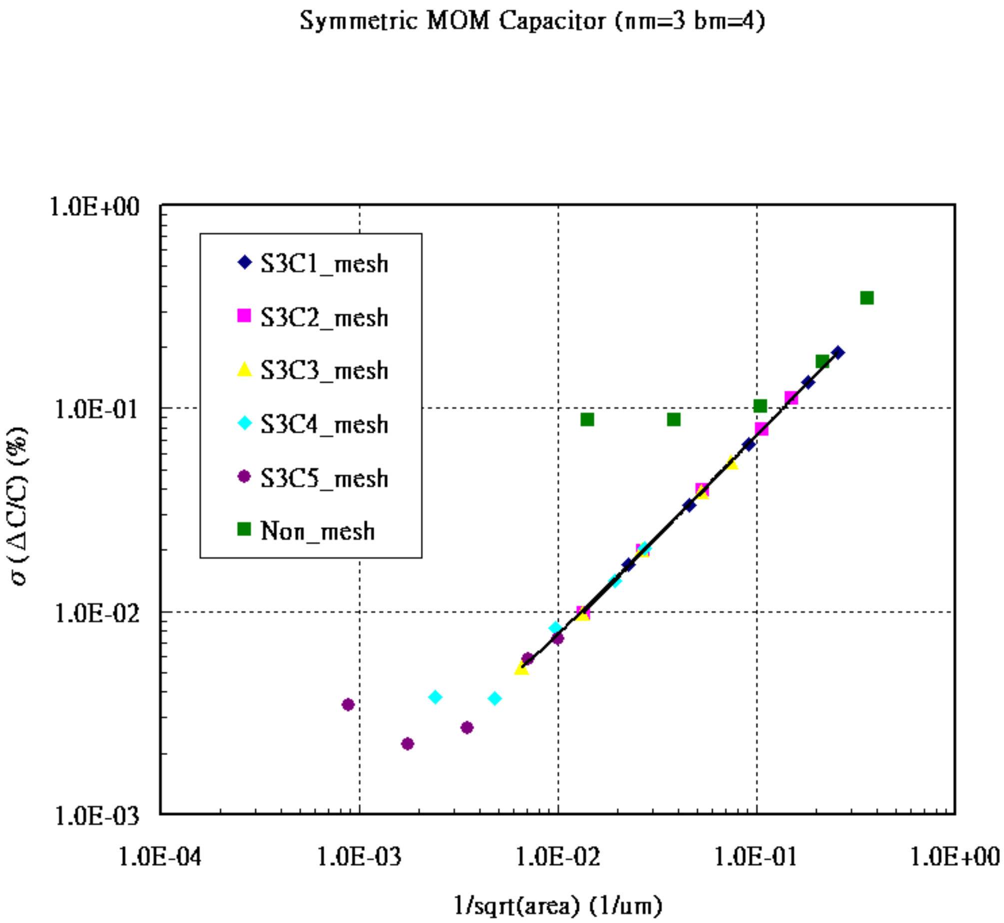

Figure 3-10 Mis-matching simulation of Symmetric MOM capacitor (nm=3 bm=4)

The following figures show the Monte Carlo mis-matching simulation results of Asymmetric MOM capacitor at 25 ℃.

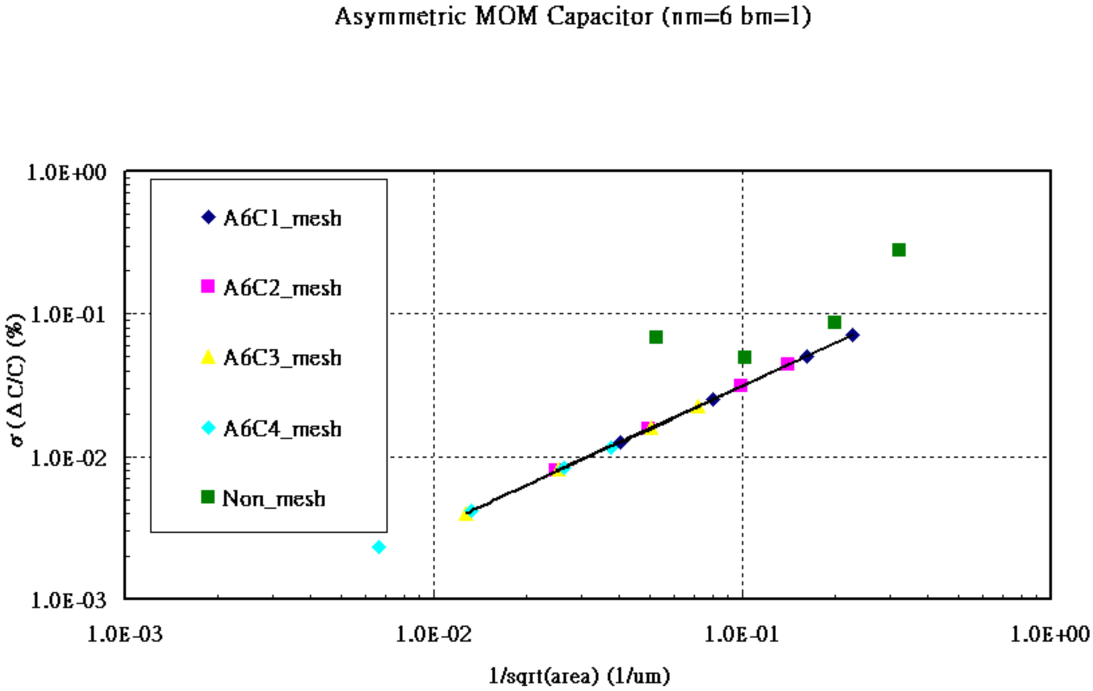

Figure 3-11 Mis-matching simulation of Asymmetric MOM capacitor (nm=6 bm=1)

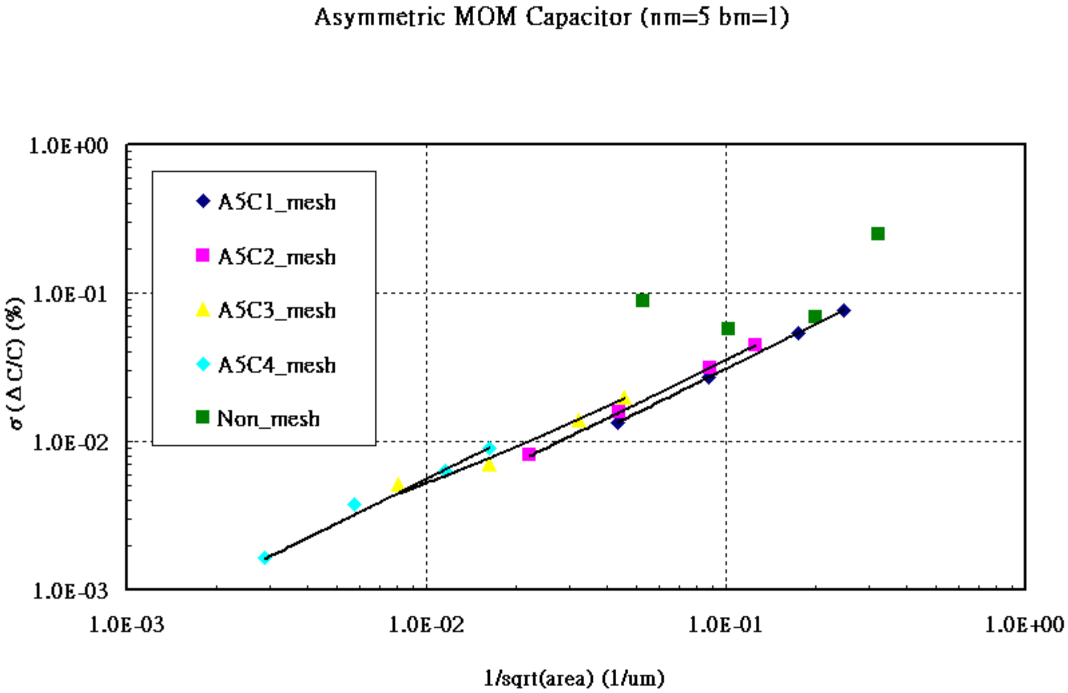

Figure 3-12 Mis-matching simulation of Asymmetric MOM capacitor (nm=5 bm=1)

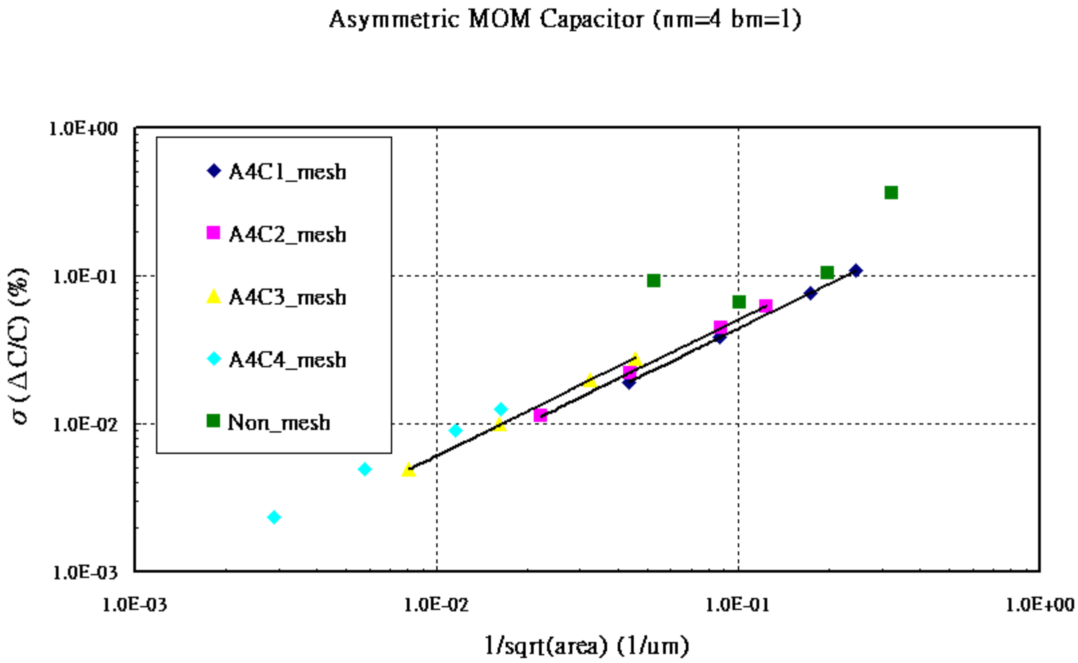

Figure 3-13 Mis-matching simulation of Asymmetric MOM capacitor (nm=4 bm=1)

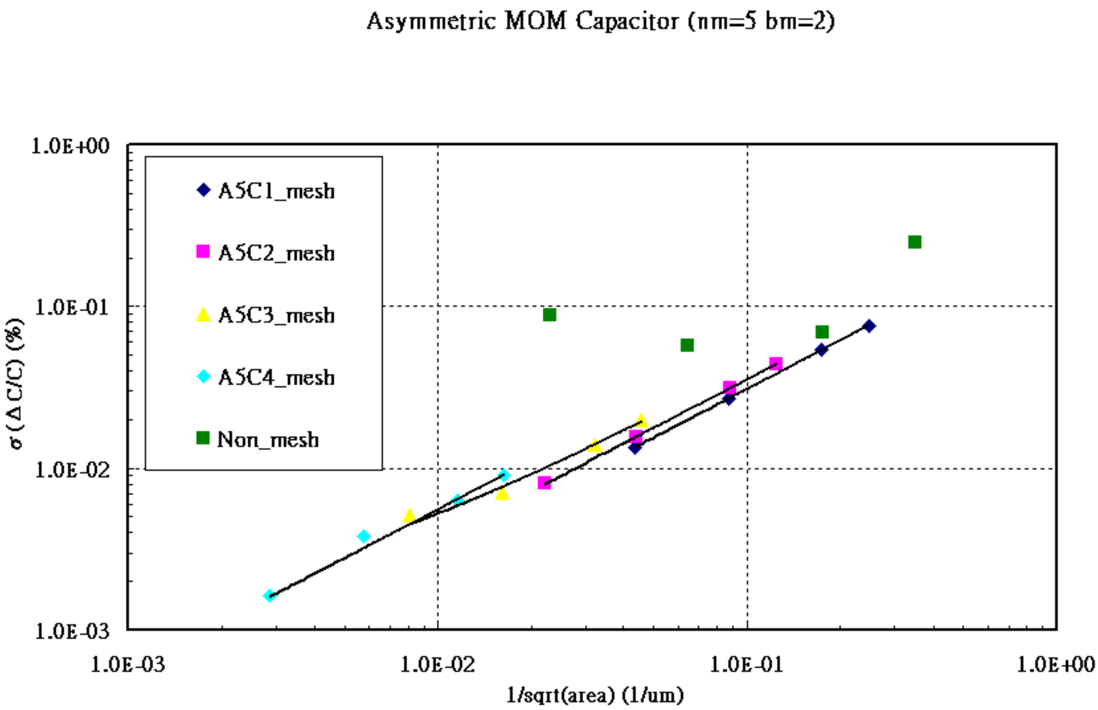

Figure 3-14Mis-matching simulation of Asymmetric MOM capacitor (nm=5 bm=2)

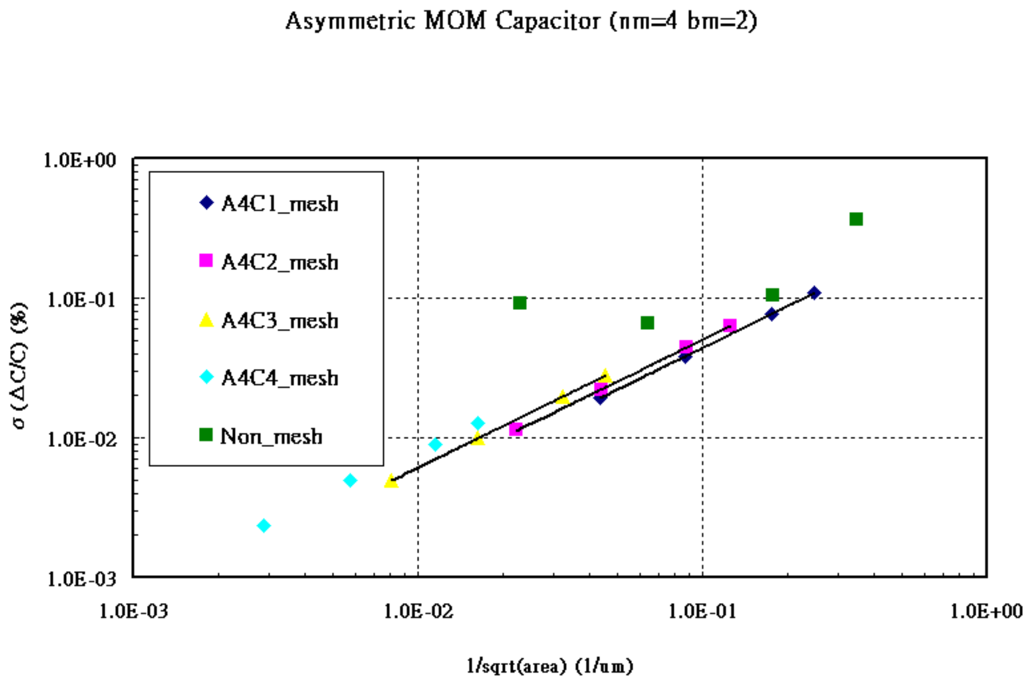

Figure 3-15 Mis-matching simulation of Asymmetric MOM capacitor (nm=4 bm=2)

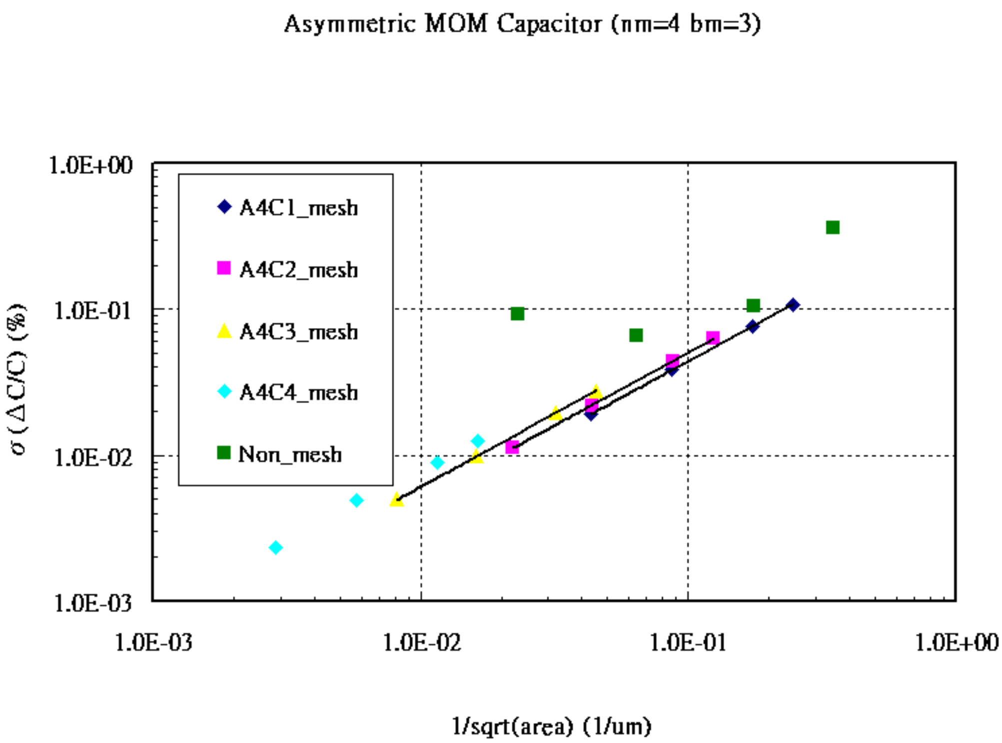

Figure 3-16 Mis-matching simulation of Asymmetric MOM capacitor (nm=4 bm=3)

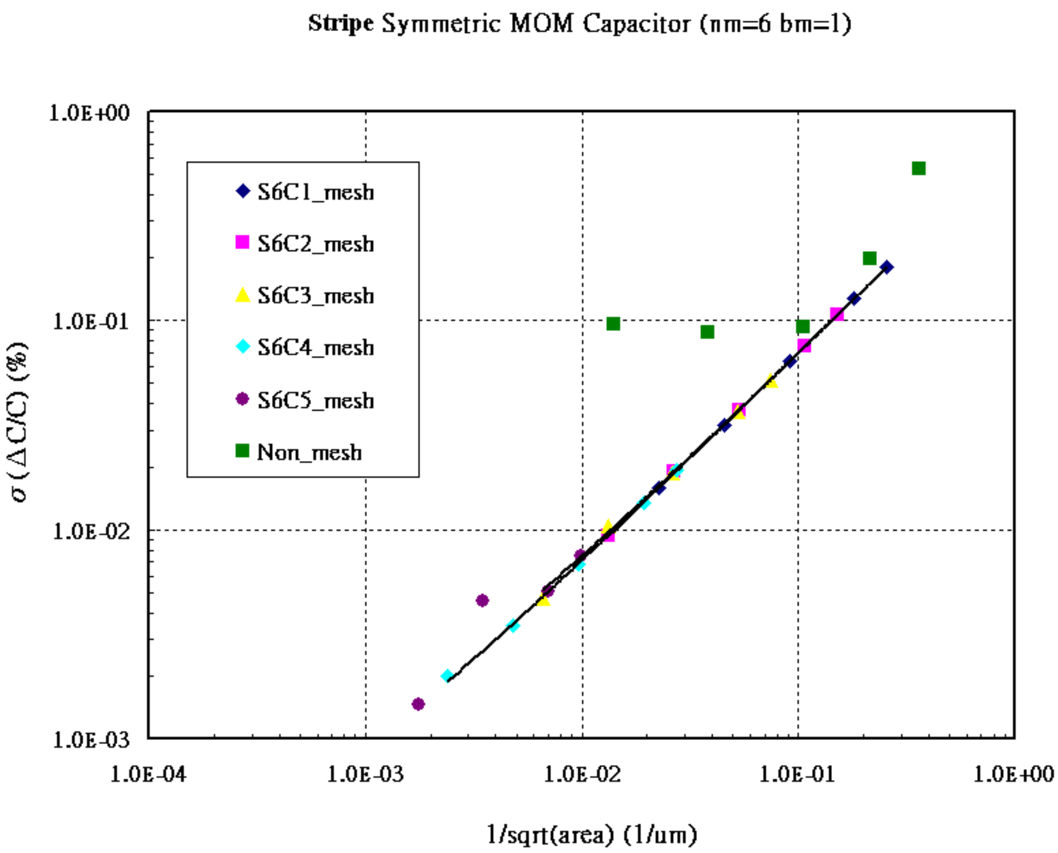

Figure 3-17 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=6 bm=1)

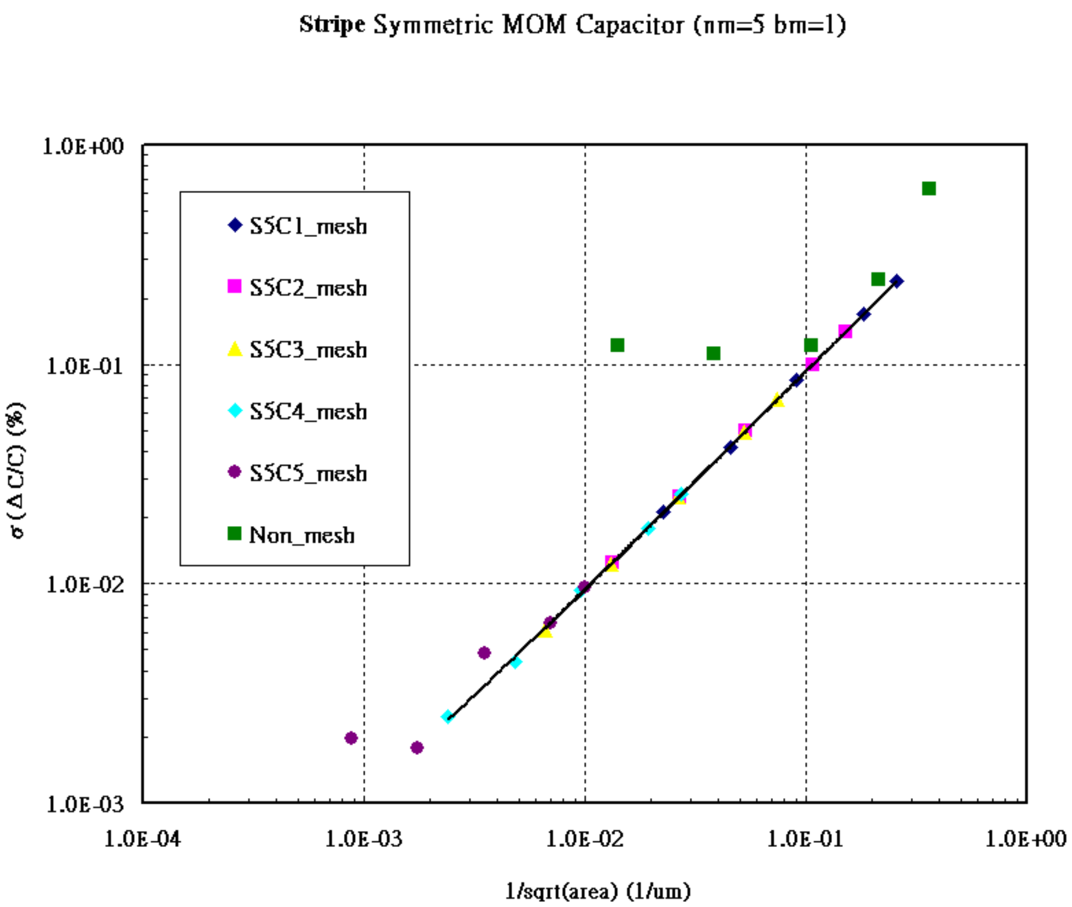

Figure 3-18 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=5 bm=1)

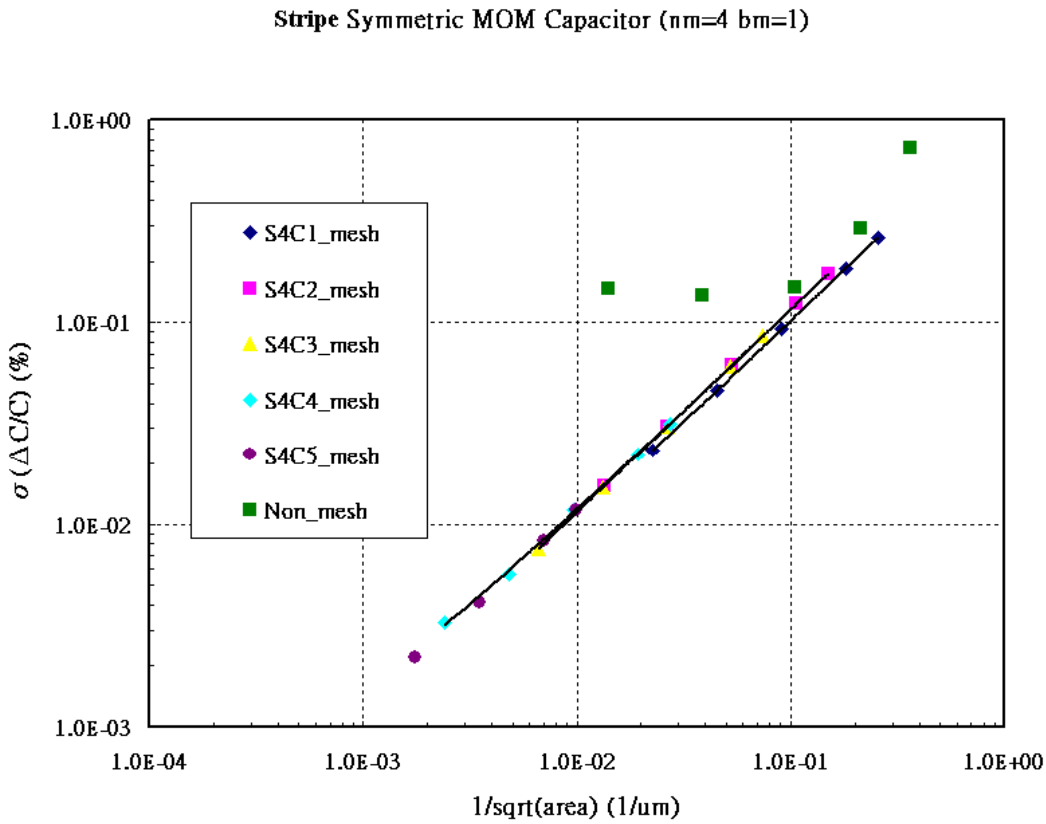

Figure 3-19 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=4 bm=1)

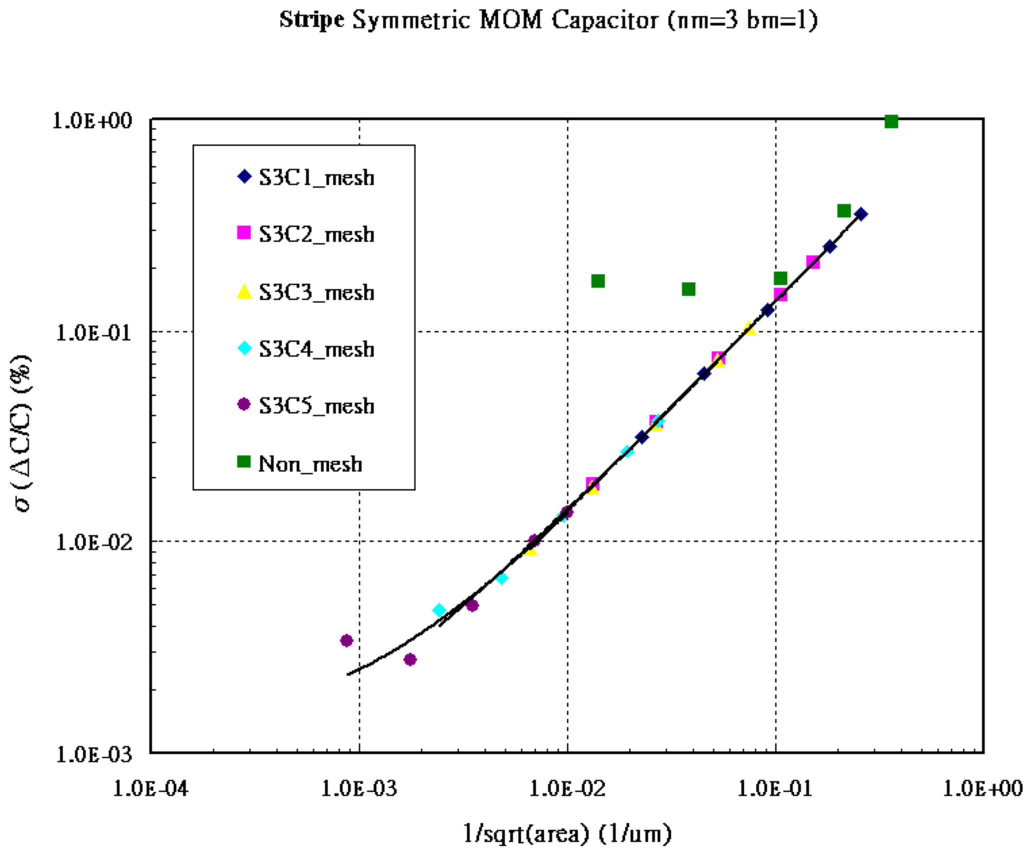

Figure 3-20 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=3 bm=1)

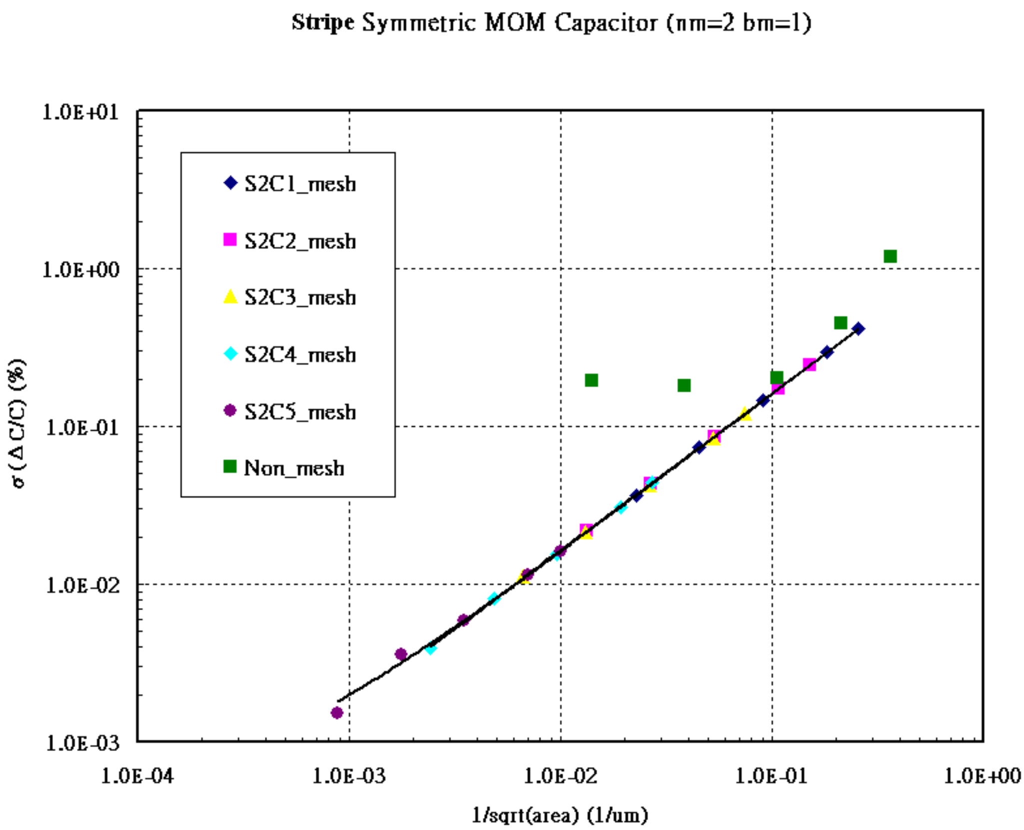

Figure 3-21 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=2 bm=1)

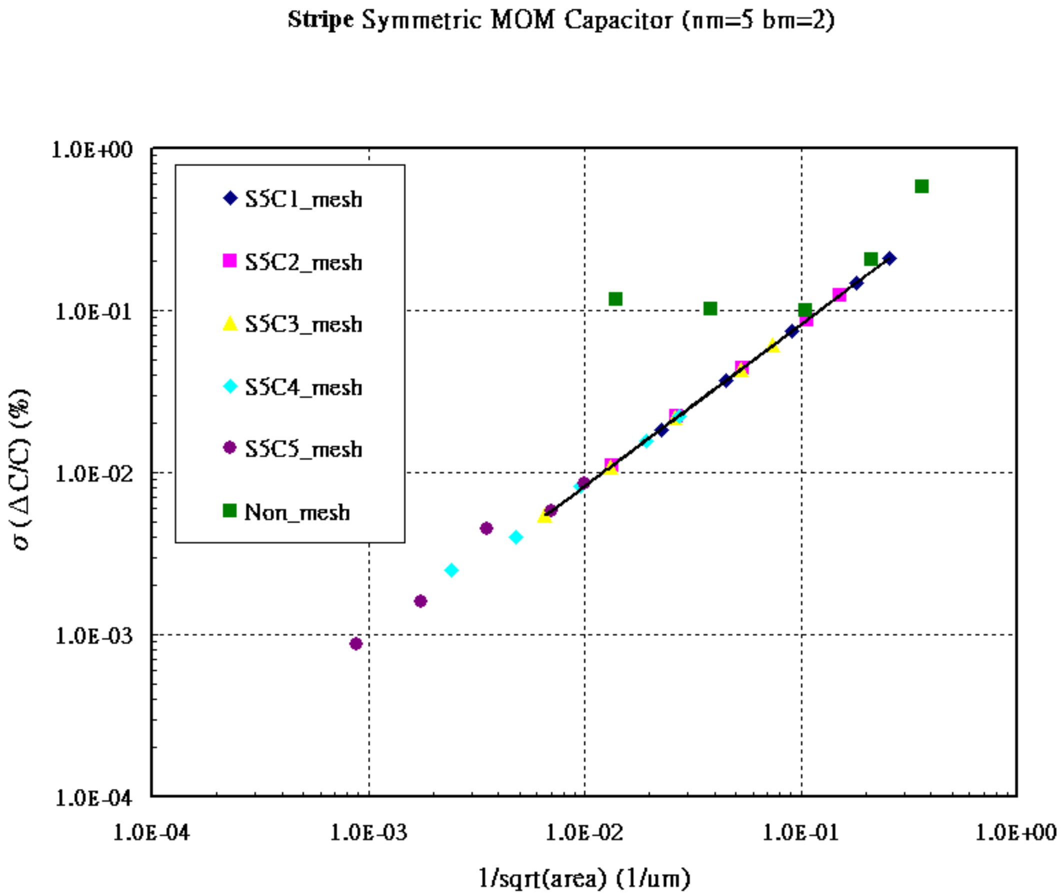

Figure 3-22 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=5 bm=2)

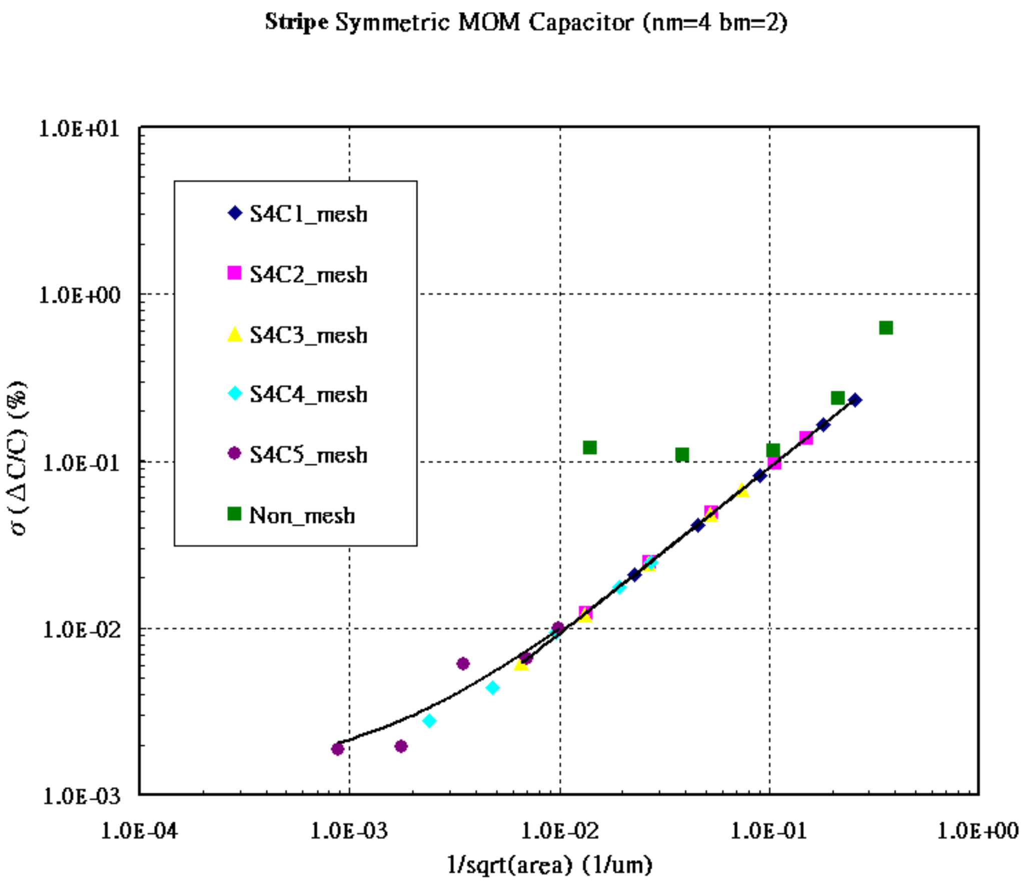

Figure 3-23 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=4 bm=2)

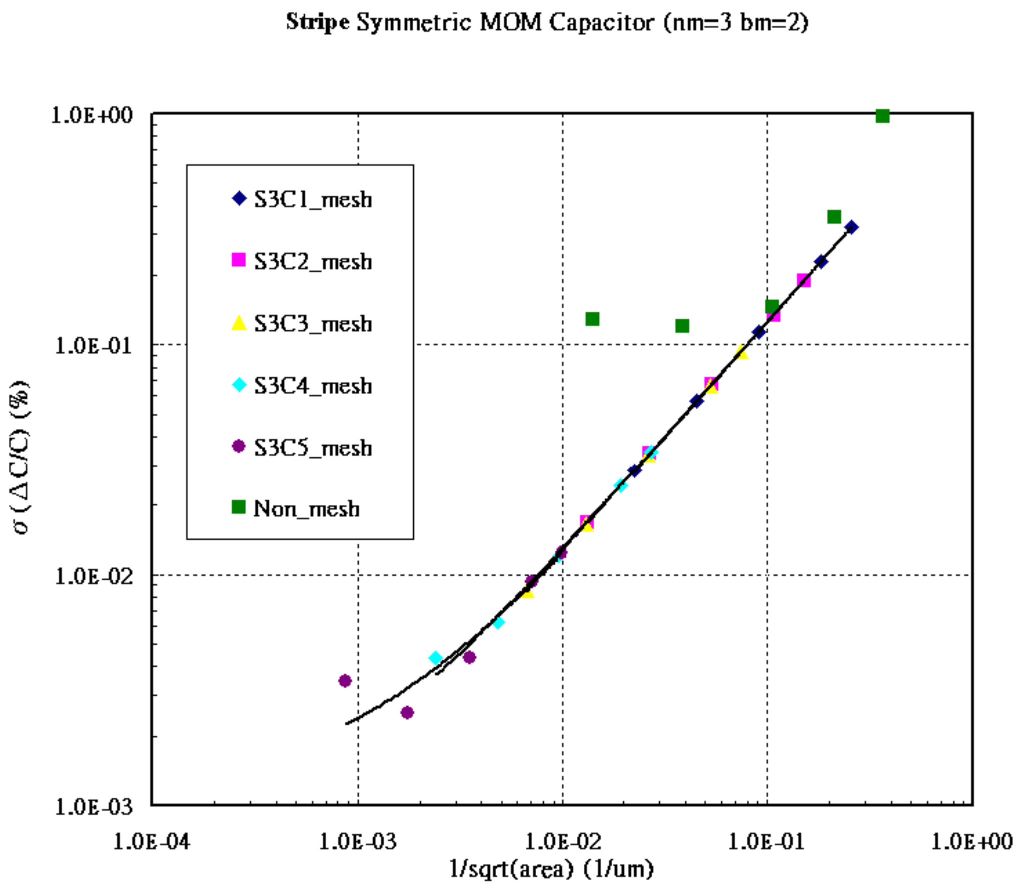

Figure 3-24 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=3 bm=2)

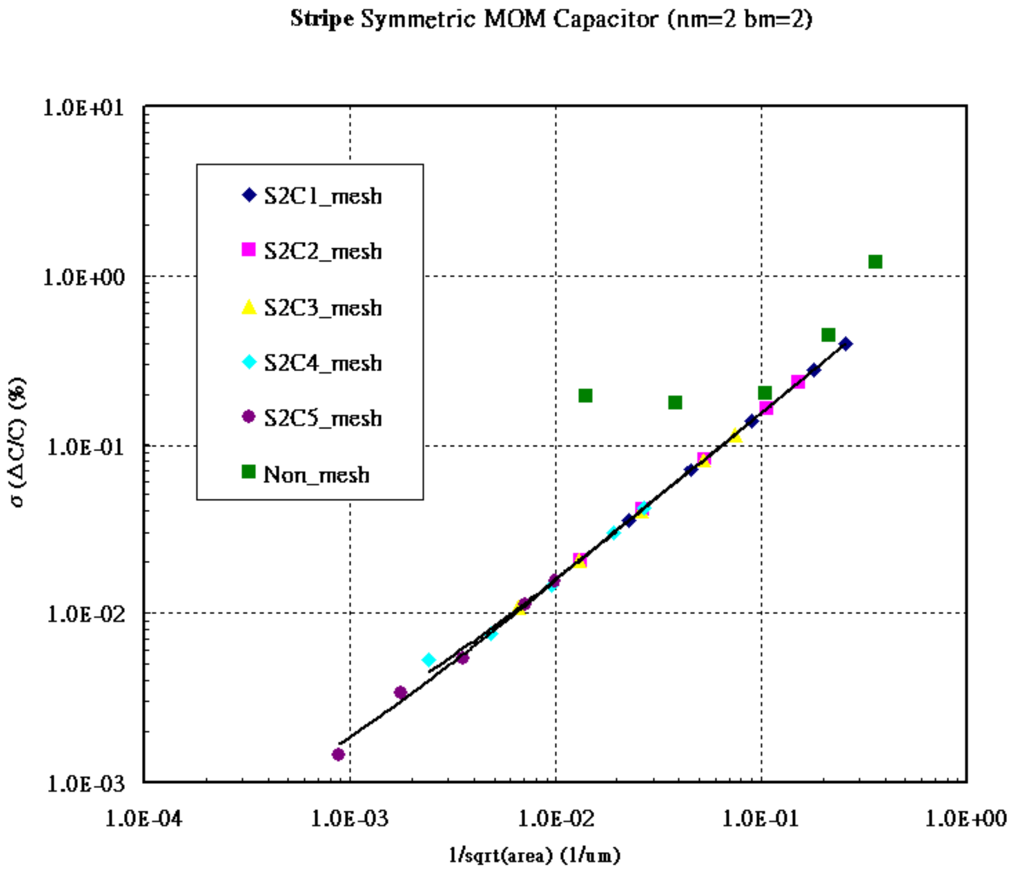

Figure 3-25 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=2 bm=2)

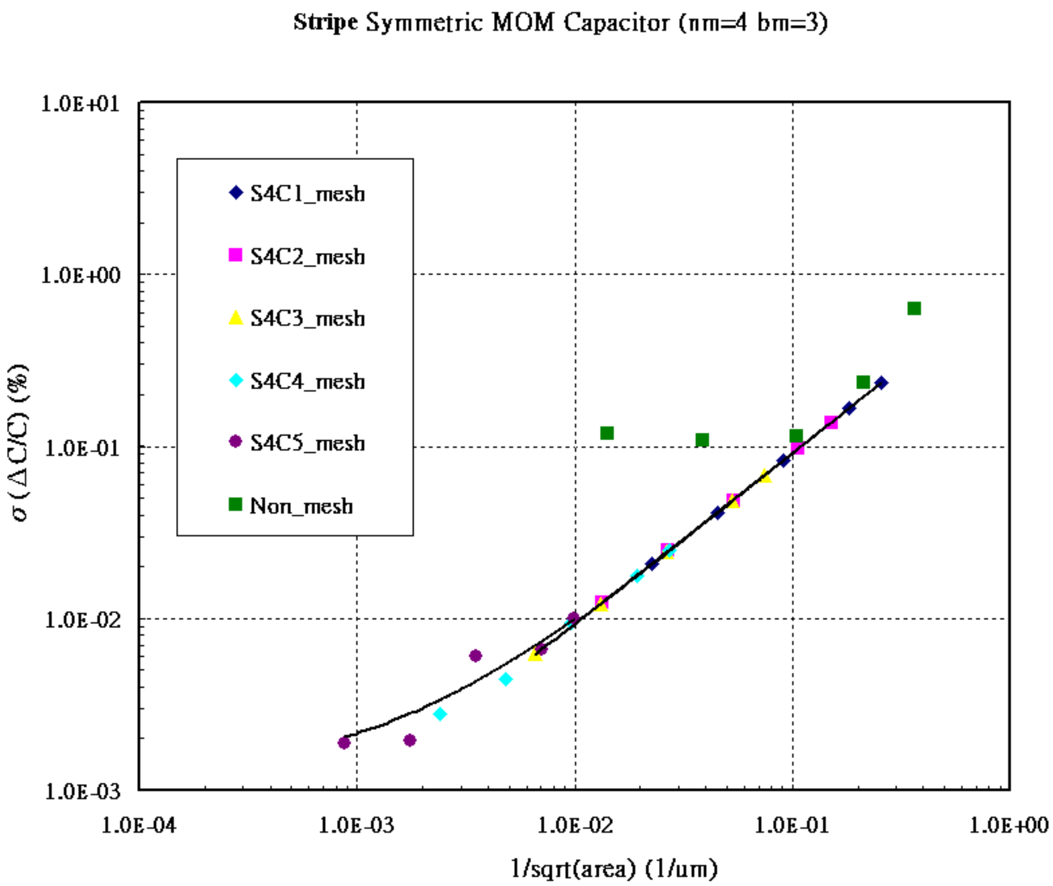

Figure 3-26 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=4 bm=3)

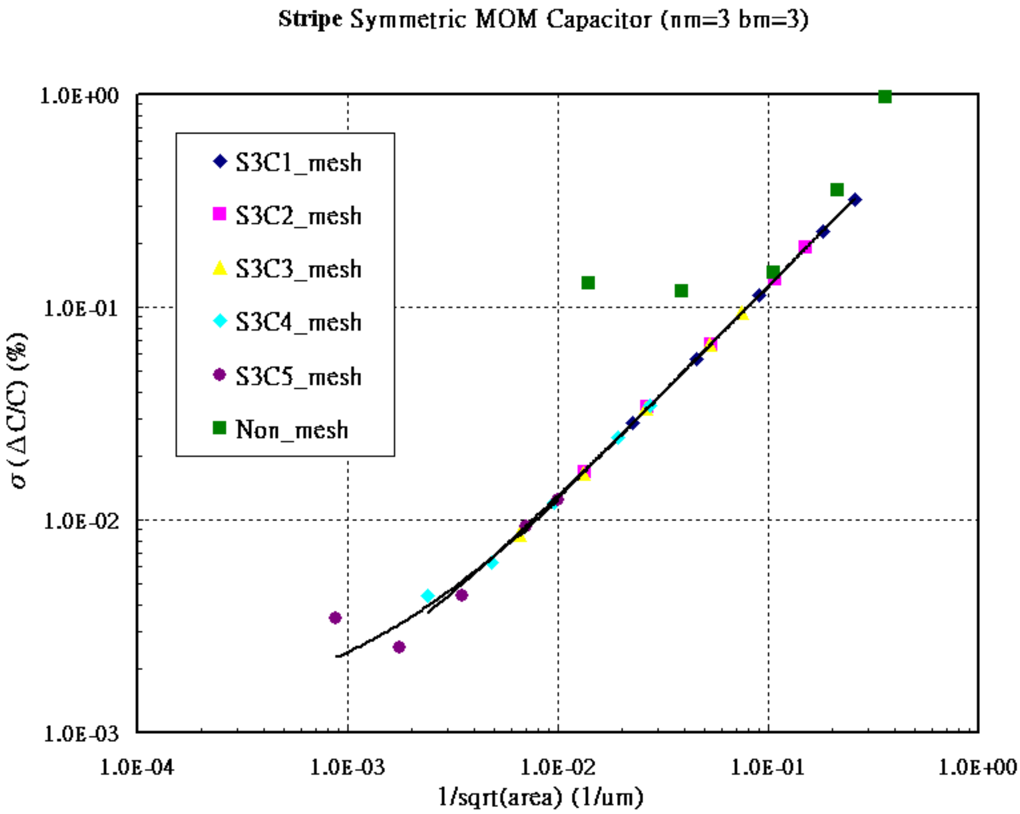

Figure 3-27 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=3 bm=3)

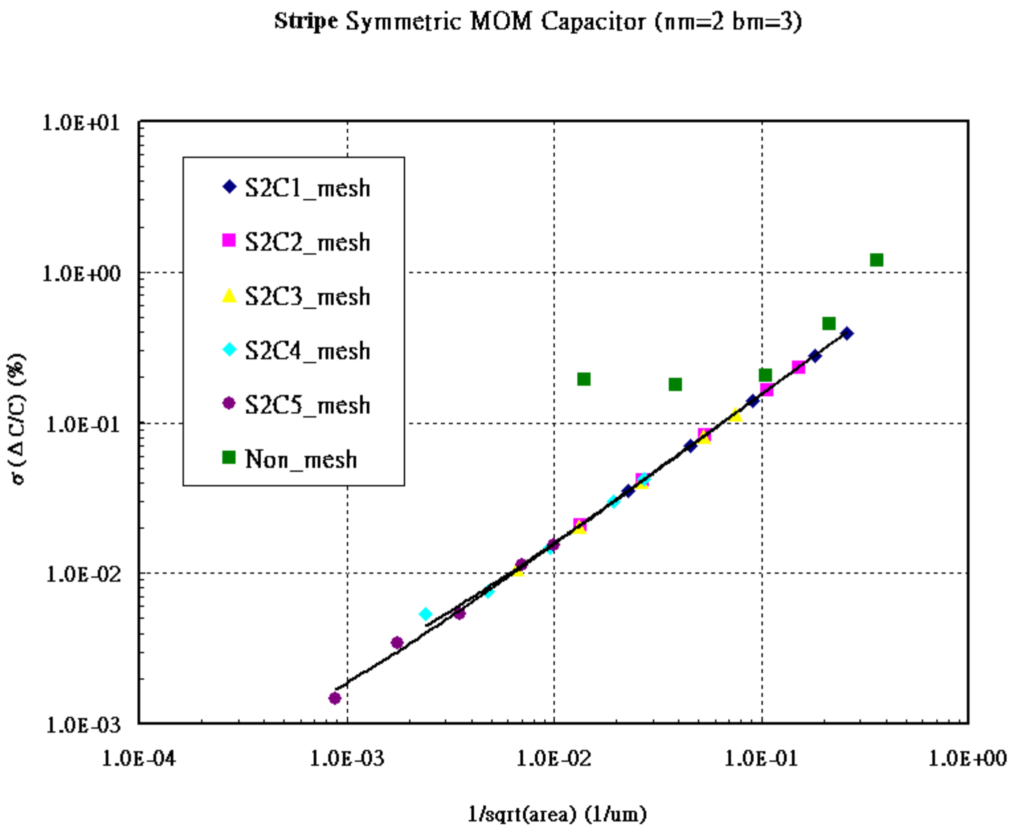

Figure 3-28 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=2 bm=3)

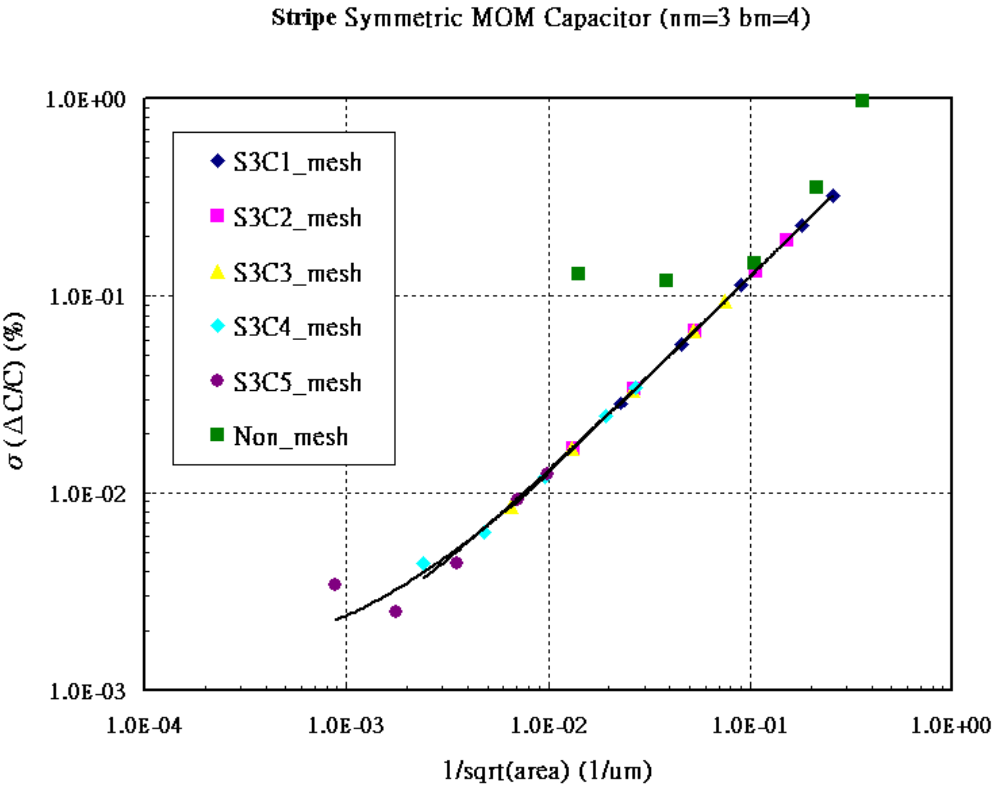

Figure 3-29 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=3 bm=4)

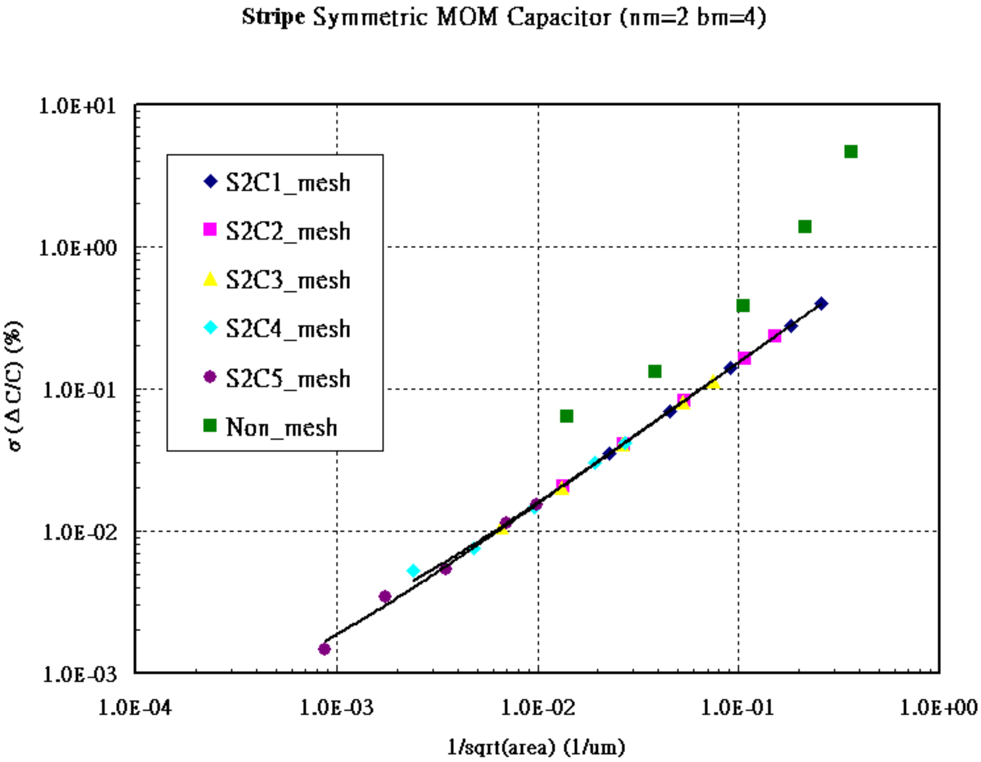

Figure 3-30 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=2 bm=4)

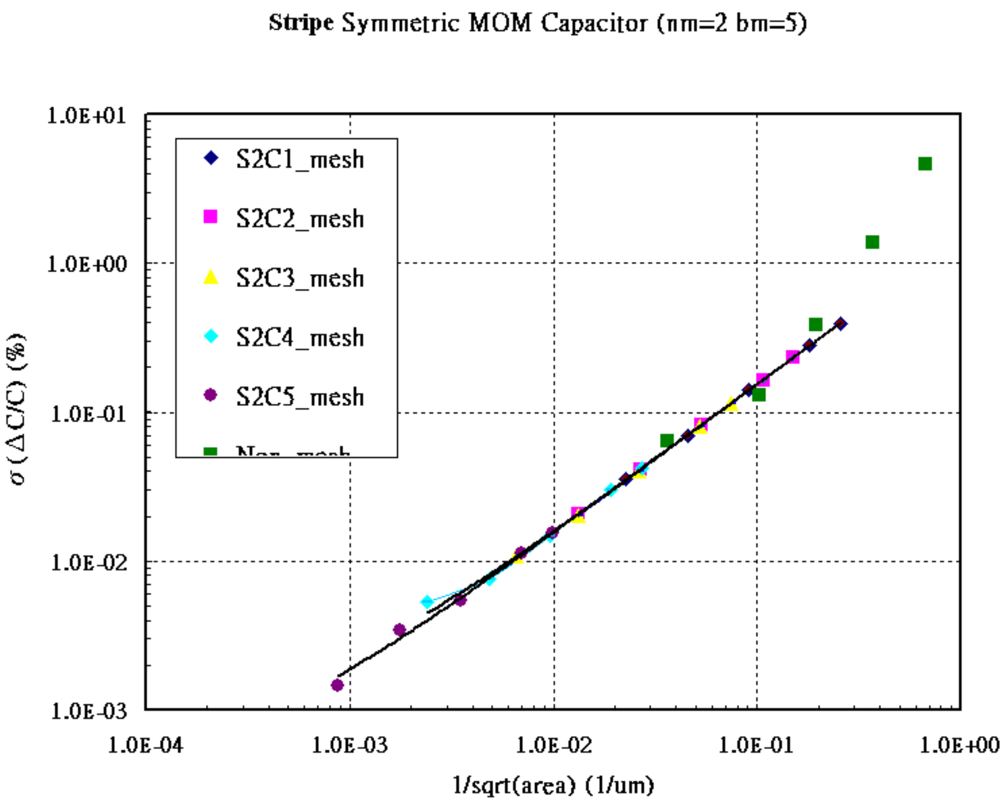

Figure 3-31 Mis-matching simulation of Stripe Symmetric MOM capacitor (nm=2 bm=5)

4 Appendix

4.1 HSPICE Warning Message

Table 4-1 Hspice Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.2 ELDO Warning Message

Table 4-2 Eldo Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.3 SPECTRE Warning Message

Table 4-3 Spectre Warning Message

NO.	Warning Message	Explanation
1.	None	None